Acting on actuation

Edited by

Hendrik De Smet

Guglielmo Inglese

Malte Rosemeyer

Conceptual Foundations of
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Chapter 1

Acting on actuation: Why here, why now?

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In their 1968 "Empirical foundations for a theory of language change", Weinreich et al. formulated a number of questions that a theory of language change should be able to answer. These include (i) the *constraints problem*, which involves delimiting "the set of possible changes and possible conditions for change" (Weinreich et al. 1968: 183); (ii) the *transition problem*, which comes down to describing the intervening stages by which "structure A evolved into structure B", as well as how features transfer "from one speaker to another" (Weinreich et al. 1968: 184); (iii) the *embedding problem*, addressing the question how language changes are embedded within individual grammatical systems as well as in specific social contexts; and (iv) the *evaluation problem*, which involves describing the subjective social awareness that speakers may have regarding ongoing changes. These four problems frame language change as shaped by factors that might be either internal or external to the linguistic system, and recognize that any given language change is deeply rooted in its social and cultural context. Addressing the four problems constitutes a necessary step to approach what Weinreich et al. consider the ultimate challenge of historical linguistics, the *actuation problem*.

Hendrik De Smet, Guglielmo Inglese & Malte Rosemeyer. 2025. Acting on actuation: Why here, why now? In Hendrik De Smet, Guglielmo Inglese & Malte Rosemeyer (eds.), *Acting on actuation*, 1–13. Berlin: Language Science Press. DOI: ??

This problem comes down to the most basic question we can ask about language change:

> Why do changes in a structural feature take place in a particular language at a given time, but not in other languages with the same feature, or in the same language at other times? This *actuation problem* can be regarded as the very heart of the matter. (Weinreich et al. 1968: 102)

Weinreich et al.'s ideas anticipated much of what was to come in the study of language change. Indeed, it is fair to say that subsequent work has been hugely successful in executing much of their program, especially with respect to the preliminary questions in (i) to (iv). For example, the constraints problem has been prominently addressed within the framework of grammaticalization studies. Continuing debate notwithstanding, many will accept that some lexical items are especially prone to grammaticalize (Bybee et al. 1994b, Kuteva et al. 2019), that some semantic changes are likelier than others (Traugott & Dasher 2001), or that many of the changes involved in grammaticalization are irreversible (Haspelmath 1999). Similarly, advances in the study of language contact have shown that constraints exist on what linguistic traits can be transmitted across communities. For instance, many will accept that nouns are more borrowable than verbs (Matras 2009), or that the intensity of contact co-determines the likelihood of structural transfer (Thomason & Kaufman 1988, Thomason 2001). Regarding the embedding and evaluation problems, research has dug into the viability of notions such as recipient design (Bell 1984) for the description of social parameters governing and resulting from language change (Mazzola et al. 2022).

Such advances raise the question whether the discipline might be edging closer to addressing its most central question. Several recent reviews (Walkden 2017, Yu 2023, Breitbarth 2024) advise caution. As Yu puts it, "a comprehensive model of linguistic change actuation […] remains elusive" (Yu 2023: 227). Similarly, Walkden concludes that "the 'actuation riddle', if solvable at all, certainly has not yet been solved" (Walkden 2017: xxx). As explained below, this is with good reason, and echoes Weinreich et al.'s own deep skepticism about the feasibility of a purely deterministic theory of language change:

> In its strong form, the theory would predict, from a description of a language state at some moment in time, the course of development which that language would undergo within a specified interval. Few practicing historians of language would be rash enough to claim that such a theory is possible. (Weinreich et al. 1968: 99)

Yet, as we want to argue here, there is also good reason to keep our eyes firmly on the long-term goal of the discipline, which in at least some ways may be less elusive than we like to think.

1 Cause for despair

For any given historical change, the actuation problem is a problem of explanation: why is this particular change occurring here and now? But the actuation problem is also a problem of prediction: given the here and now, what change is going to occur? A solution to the actuation problem would amount to a satisfactory causal explanation of a given change or a correct prediction of change to come. So, why is that so difficult to achieve?

Two centuries[1] of systematic historical linguistic research will have driven home the point that language change is complex. To begin with, historical linguists are generally aware that any specific change may plausibly involve a multiplicity of causes. For many changes, the research literature offers a range of possible explanations. For example, the rise of the periphrastic HAVE perfect in Romance to mark perfective aspect, as in (1), has been attributed to grammaticalization (Harris 1982), systemic constraints within the domain of temporal auxiliation (Pountain 1985), and the influence of the Carolingian reforms on literate scripture, leading to areality (Drinka 2017).

(1) a. J'ai décidé de changer de voiture
'I've decided to change my car'

b. el mismo efecto se ha notado en los resultados de las niñas en matemáticas
'the same effect has been observed in girls' results for mathematics'

It would be a mistake to believe that only one such explanation can be correct for any one change. Joseph (2013: 675) proposes that

> If we can identify multiple pressures on some part of a language system, it cannot always readily be excluded that some or even all might have played a role in shaping a particular development; if all of the factors represent reasonable pressures that speakers could have been aware of and influenced by, excluding any could simply be arbitrary.

[1] No doubt an arbitrary cut-off, but Grimm's Law was formulated (at least by Grimm) in 1822.

Sometimes multiple causation is obvious. Consider, for example, the emergence of intrusive *r* in non-rhotic varieties of English (as when *draw a circle* is pronounced /dɹɔːɹəsɜːkəl/). Although the exact mechanism is disputed (Hay & Maclagan 2012), it is perfectly plausible that the change is, among other things, *both* a consequence of the loss of rhoticity, *and* of a natural tendency for hiatus avoidance that in turn stems from the physiologically imposed "jaw cycle constraint" (Redford & van Donkelaar 2004, Cox et al. 2014). Multiple causation of course seriously complicates any attempt at explanation, let alone prediction.

More generally, on the internal side, language is a hugely complex system that can recruit a variety of resources to achieve an equally great variety of communicative goals. For that reason, it has been termed a complex-adaptive system (Van de Velde 2014). Unsurprisingly, then, it is hard to predict how such a system will respond to any new situation. After all, from the perspective of language users, any communicative problem that arises may have a variety of solutions. Conversely, any communicative solution selected will be a compromise between a variety of potentially conflicting demands. There is little doubt that it is this level of complexity that makes therapeutic explanations of language change so fraught (Reinöhl & Himmelmann 2017). Again, complexity will defy any attempt at easy explanation or prediction.

Finally, among these multiple causes of language change, there are also language-external causes. These essentially stem from outside the realm of what can be linguistically modelled, reflecting rather the contingencies of history, which continually upset the ecology in which language functions. They include demographic, cultural and technological developments leading to the continuous emergence of new contact situations, to ever-shifting relations within speech communities, to changing attitudes towards language, to changes in how language is transmitted, and so on. While linguists can obviously study the impact of such external developments on language, they can hardly be expected to predict the developments themselves. Weinreich et al., too, had pointed to the unpredictability of social behaviour as a major obstacle to solving the actuation riddle:

> If we seriously consider the proposition that linguistic change is change in social behavior, then we should not be surprised that predictive hypotheses are not readily available, for this is a problem common to all studies of social behavior. (Weinreich et al. 1968: 186)

None of this makes addressing the actuation problem any easier. What is more, these fundamental difficulties are compounded by additional uncertainties. The seemingly trivial matter of dating a change, for instance, can be exasperatingly difficult. In fact, even agreeing on what exactly counts as change proves difficult. Changes presuppose innovation, whereby a speaker produces a linguistic behaviour that has no precedence in their linguistic experience. But does innovation constitute a phenomenon in its own right that can be easily distinguished from the social propagation of change or not (De Smet 2020)? And how do innovations relate to synchronic productivity – or, for that matter, to speech errors? More disconcertingly still, linguists continue to struggle to find agreement on several fundamental questions about the nature of the language system and its cognitive underpinnings. On an entirely practical note, the time-scale on which language change unfolds tends to make it difficult to test any predictions, or replicate change experimentally. And then there is the historical record, which is never as complete as we would want it to be. But does all of that really make things hopeless?

2 Cause for hope

When historical linguists are not getting bogged down by intractable problems, they do in fact advance the discipline. A number of developments have been bringing solutions to the actuation problem within closer reach. These include both conceptual and empirical advances.

2.1 Conceptual advances

On the conceptual side, as Weinreich et al. argued, "long before predictive theories of language change can be attempted, it will be necessary to learn to see language… as an object possessing orderly heterogeneity" 1968: 100. Since then, the study of linguistic variation has become a core component of the discipline. Linguists have learned to see variation as intrinsic to the language system and to individuals' linguistic knowledge. This, in turn, has engendered a new mode of thinking about language.

First, variation has forced linguists to think about the choices speakers make – that is, how speakers decide what to say and how to say it. This focus on the process of linguistic selection has increasingly come to take center stage in general models of language and language change (Haspelmath 1999, Croft 2000, Beckner et al. 2009, Berg 2014, Enfield 2014, Schmid 2020). What these models have in common is that they use linguistic selection as the unifying concept that can tie together the various external and internal forces exerting their influence on language structure. Moreover, through their focus on selection, these models can in principle naturally incorporate both the innovations by individual speakers, and the propagation of innovations within (or across) linguistic communities. If we can understand the choices speakers make, we understand why language is organized the way it is, and why it changes the way it does. It is in speakers' choices that all the complexities of the actuation problem come together. The fact that linguists are seriously contemplating linguistic selection is therefore a hopeful sign.

Second, among external forces behind language change, a key role has been attributed to language contact, already by Weinreich et al. (1968). However, it is only in the last decades that more accurate models of change in contact situations have been elaborated, so that we now have a much better understanding of the factors that constrain contact-induced change, in terms of what can be borrowed and under which linguistic and social circumstances (e.g. Thomason & Kaufman 1988, Matras & Sakel 2007, Gardani 2020, Sinnemäki et al. 2024, see Walkden 2017: 415–417). Such advances may give us an increasingly firmer grip on what are possible consequences of the external factors that play a role in language change.

Third, variation has opened linguistics to probabilistic thinking. Not only has it become clear that speaker choices can only be described and understood probabilistically, speaker's mental representations of language actually code information about frequencies and likelihoods (Plunkett & Marchman 1993, Bod 2015), and frequency of usage plays a key role in language change at large, especially in terms of automatization of physiological gestures and cognitive sequences (Bybee 2015). As there can be little doubt that a predictive theory of language change will be a probabilistic theory, linguists' shift towards probabilistic thinking is another hopeful sign.

2.2 Empirical advances

With regard to a "fully worked-out theory of linguistic change", Weinreich et al. doubted "whether any linguist has enough relevant facts at [their] disposal to attempt anything so ambitious" (1968: 102), thus echoing Bloomfield's 1933: 374 skeptical observation that "language change has never been observed." On that score too, however, the discipline has made obvious progress.

First, language documentation efforts are increasingly capturing the diversity of the world's languages, thereby also paving the way for ever more robust typological generalizations. Among other things, this has given historical linguists a much firmer basis to recognize regularities of language change. Such regularities include grammaticalization (Bybee et al. 1994a, Hopper & Traugott 2003), subjectification (Traugott & Dasher 2001), and cyclical change (van Gelderen 2008). Each of those regularities is in itself a probabilistic prediction about language change, but what is more, they also give historical linguists the kind of baseline they need to assess the likelihood of a given type of change and begin to work out the conditions that promote or prevent its occurrence (Rosemeyer & Grossman 2021).

Second, the discipline has seen a surge in the availability of usage-data, especially in the form of corpora. For some languages and some historical periods, the readily available usage-data is now so rich that change can be observed and described in extreme detail, down to the level of the smallest qualitative steps in individual language users (Petré & Anthonissen 2020). With more and more varied usage-data available, more opportunities open up to generate and test hypotheses about the mechanisms that drive language change (De Smet 2016, Simonenko et al. 2019). A happy side-effect, moreover, is that change is increasingly being studied on shorter time-scales, which may open the way towards predictions that are realistically testable. Finally, recent studies have demonstrated the necessity and possibility to assess the extent to which written data might represent change occurring in natural, informal speech at the point of creation of texts, thus disentangling environmental and grammatical change (Szmrecsanyi 2016, Rosemeyer 2019).

Third, the development of new stochastic models for the study of populations and networks, such as agent-based models, has provided new promising tools to investigate how language change propagates within communities. In this respect, agent-based models have offered new insights on both sound change (see discussion in Yu 2023) and morphological change (Dekker et al. 2022).

In short, there are reasons to be hopeful. Several developments in the discipline are bringing us closer to solving the actuation problem. That there is hope is particularly evident from recent work that directly confronts the actuation problem. This is true especially of sound change (see e.g. Baker 2008, Stevens & Harrington 2014, Pinget 2015, Yu 2013, Hall-Lew et al. 2021, see also Yu 2023 for a recent overview). With regards to morphosyntactic change, too, some have ventured forward (see e.g. Cheshire et al. 2013, Backus 2015).

3 Can historical linguistics be a predictive discipline?

When thinking about the actuation problem (and its possible resolution), we must keep in mind that the explanatory and predictive power of a scientific model are always a matter of degree. Climate scientists do not dismiss their models because they could not predict a specific draught, or a single particularly wet summer. If exact prediction is the goal, then that goal is usually either trivial or hopeless. But if the goal is to advance the exactness of prediction, then it is worthwhile and achievable. In the same vein we believe solving the actuation problem is a matter of degree – an ideal target the discipline can and should strive to incrementally approximate.

As it stands, historical linguistics as a discipline seems divided as regards the question of whether historical linguistics can and should be a predictive discipline. In July 2022, right before the workshop ”Acting on actuation” at the International Conference of Historical Linguistics 2022 in Oxford, which served as the starting point for the present volume, we conducted a non-representative survey among 22 colleagues from historical linguistics. One of the questions was ”Can historical linguistics be a predictive science?”. A majority of the colleagues argued that this is impossible. For instance, one colleague claimed that

> [The actuation problem] addresses the ontological question of explainability/predictability in (historical) linguistics, one we cannot resolve in the same terms as the nomological-deductive ’hard sciences’, hence signals our discipline’s border limit.

Another colleague argued that:

> If we could quite generally solve the actuation problem, a strongly predictive theory of language change in the future could be within our reach. (I am not optimistic, though.).

Despite these skeptical voices, our goal is to persuade the historical linguistic community that the actuation problem is not unassailable. To that end, in this volume we have collected a number of papers from researchers with different background and expertise to jointly tackle the most basic question about language change.

In their paper “Contact-induced language change and its constraints in relation to actuation”, **John Hawkins and Luna Filipović** rethink actuation from the perspective of bilingualism research. They develop a multi-factor complex adaptive systems approach to language change, which allows modelling the likelihood of any change to occur in terms of the ranking of a series of principles of language change. In a similar manner to Hawkins and Filipović, **Henri Kauhanen**’s contribution “What can toys teach us about actuation?” proposes a model-based solution for studying actuation. His application of the AB toy, a simplified model of competition between two linguistic variants, points to the crucial role of social factors and raises questions about what exactly constitutes a solution to the actuation problem. The paper "Parallel linguistic and sociocultural change: Modals and mores", by **Rena Torres Cacoullos and Catherine Travis**, explores the potential of historical social variationist linguistics in explaining actuation. Their diachronic analysis of Australian English modals shows that the rise of need to is intertwined with a more general sociocultural change in the concept of obligation. **Anne-France Pinget**'s contribution "Sound change in present-day Dutch: A variationist, synchronic approach to the actuation problem" studies an ongoing sound change in progress, devoicing of labiodental fricatives in Dutch. Her results support a definition of actuation as an incremental accumulation of an innovative property in an individual speaker's system. The paper "Relative clause-forming strategies in German: On typological generalizations and diachronic language change" by **Ann-Marie Moser** demonstrates the strength of comparative analyses for attempts to study actuation. Moser investigates historical change in relative complementizers in different German dialectal areas, explaining differences in these historical processes through language-internal factors.

These studies represent crucial advances towards solving the actuation problem both from a methodological and theoretical perspective. Several contributions demonstrate the need for a model-based approach towards actuation; in order to successfully explain why change comes about it is necessary to also explain variation. They show how these models can incorporate both individual and social determinants for change. Finally, they use comparative approaches (between age cohorts and dialects) in order to address the question of non-change. In doing so, these papers develop fresh theoretical perspectives on the actuation problem. These involve discussions of the relationship between different principles of language change, the role of contact and bilingualism, the distinction between innovation and propagation, and the role of sociocultural change. To summarize, papers collected in this volume act on actuation and demonstrate how modern historical linguistics can successfully address the actuation problem.

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Chapter 2

Contact-induced language change and its constraints in relation to actuation

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The central problem for language change that has been formulated as the "actuation problem" (Weinreich et al. 1968) involves explaining why changes happen when they do and in the languages they do, but not at other times or in other typologically similar languages. One factor that has emerged as a "cause for hope" in solving the problem has been language contact: a particular change takes place in a given language at the time it does because this language is in contact with another language from which the property in question has been borrowed. In this paper we argue that this cause for hope needs to be considerably refined if it is to be helpful in relation to the actuation problem. The crucial refinement we need is to talk about bilingualism, and specifically different types of bilingualism sociolinguistically and psycholinguistically and with different pairs of languages, in order to better understand whether there will be any influence from one on the other. Even if the type of bilingualism and its distribution throughout a speech community at a given time does lend itself to borrowing, the actuation of any change will be further constrained by general grammatical and universal principles on the one hand, and by diachronic processes on the other involving gradualness and ease of innovation. This paper builds on the general model of bilingualism, CASP (short for "complex adaptive system principles") developed in Filipović (2019) and Filipović & Hawkins (2019), and applies it to language change and the actuation problem. By shifting the emphasis from contact to bilingualism we gain greater clarity on when contact does lead to change. Conversely, when changes are observed, we gain greater clarity on whether these changes were the result of contact.

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1 Introduction

The "actuation problem" in historical linguistics was formulated by Weinreich et al. (1968: 102) as follows: "Why do changes in a structural feature take place in a particular language at a given time, but not in other languages with the same feature, or in the same language at other times?" One potential solution for it that has been proposed is language contact: a particular change takes place when it does because the language in question is in contact with another language from which the property has been borrowed. And indeed language contact is increasingly seen as a general source of historical change (Weinreich 1953, Thomason 2001, Harris & Campbell 1995, Campbell 2004), and we now have numerous case studies documenting such changes (see e.g. Mithun 1999, 2010 for contact-induced changes in native North American languages, Aikhenvald 2002 for contact and change in Amazonia, and more generally the handbook of Darqueness et al. 2019 for extensive further examples from around the world).

But contact does not inevitably lead to language change. In fact, most of the properties that contrast between languages in contact at any given time are *not* transferred from one to the other. In this paper we argue that we can achieve a better understanding of what happens in contact situations, and we can make more of a contribution to the actuation problem, by looking more systematically at:

(i) bilingualism rather than contact per se, incorporating recent sociolinguistic and psycholinguistic findings and theories from the bilingualism research literature; in conjunction with

(ii) general grammatical, especially universal, constraints on current languages, plus general diachronic laws and patterns of change from historical linguistics, and the best available explanations for these universal and diachronic constraints.

To this end we draw on the integrated socio- and psycholinguistic model of bilingualism developed in Filipović (2019), Filipović & Hawkins (2019) which is referred to as "CASP", short for "complex adaptive system principles" of bilingual language learning and processing. CASP was inspired by the general notion of language as a complex adaptive system in the sense of Gell-Mann (1992) in which multiple factors interact to produce a range of observable outcomes. The CASP model draws on the now extensive literature on bilingualism in both sociolinguistics and psycholinguistics and shares its key features with other complex adaptive systems approaches to language learning and use, as exemplified in e.g. the papers in Ellis & Larsen-Freeman (2009)'s edited volume on Language as a Complex Adaptive System. These key features are defined in the position paper by Beckner et al. (2009: 2) as follows:

> (a) The system consists of multiple agents (the speakers in the speech community) interacting with one another. (b) The system is adaptive; that is, speakers' behavior is based on their past interactions, and current and past interactions together feed forward into future behavior. (c) A speaker's behavior is the consequence of competing factors ranging from perceptual mechanics to social motivations. (d) The structures of language emerge from interrelated patterns of experience, social interaction and cognitive processes.

Because CASP incorporates and defines the multiple cooperating and competing forces that are at play in bilingualism and spans both social and psychological pressures (in a field in which much of the research focuses on just one or the other) it can solve a number of puzzles and contradictory findings in the literature (see Filipović & Hawkins 2019 for a summary paper). It can also help us in the present context of language change and actuation to better understand when language contact will lead to change in one or the other language of a bilingual speaker. These changes may then spread to the wider speech community comprising monolinguals, if social and demographic circumstances favor this (see Trudgill 2011 for detailed discussion of these).

There are important further constraints on such bilingualism-induced changes, however, involving universally permitted co-occurrences of grammatical properties on the one hand, and well-attested diachronic constraints on successive language states on the other. By integrating these grammatical and diachronic principles, and proposed explanations for them, with CASP and its predictions for convergence under bilingualism, we achieve a more complete understanding of when contact does or does not lead to change. Conversely, when changes occur we achieve a better understanding of whether contact was actually responsible for them.

The order of presentation is as follows. In Section 2 we define one of CASP's central principles for changes in usage and grammar under bilingualism, Maximize Common Ground (MCG), and summarize illustrative data supporting its predictions. The constraints internal to CASP are of both a sociolinguistic and a psycholinguistic nature and are enumerated in Section 3. Section 4 discusses filters on these Maximize Common Ground predictions stemming from language universals, while Section 5 summarizes general constraints on successive diachronic states from the historical linguistics literature. Section 6 summarizes the resulting multi-factor model of bilingualism-induced language change proposed here. It is because of these multiple factors that contact per se does not necessarily lead to change at any given time and place. Section 7 concludes with the theoretical point that a focus on different types of bilingualism in their interaction with general grammatical and diachronic laws provides a helpful step towards solving the actuation problem.

2 Bilingualism and efficiency: The CASP model

The CASP model (2019) is built on general principles of efficiency in language use and learning, as summarized in recent psycholinguistic work by e.g. Gibson et al. (2019). One of CASP's major goals is to account for the many convergences among languages that have been observed in the bilingual literature, in both usage and grammatical conventions. The central principle proposed is Maximize Common Ground. CASP's other principles are shared with monolingual learning and usage, namely Minimize Learning Effort ("master shared properties between two languages first"), Minimize Processing Effort ("make use of simple properties rather than complex ones whenever possible"), Maximize Expressive Power ("master complex properties when these are needed in order to express all meanings in both languages"), and Maximize Efficiency in Communication ("use complex properties only when simple ones are not enough for the purpose"), see Filipović (2019: 56–60) for definitions and a summary.

2.1 Maximize Common Ground (MCG)

The bilingual literature tells us that bilingual speakers and learners regularly make "common ground" between their two languages: see the many examples in Filipović (2019) and in other bilingualism models and texts on borrowing and transfer such as Muysken (2000), Myers-Scotton (2003), De Groot (2011), Pavlenko (2011, 2014), Silva-Corvalán (2014), Field (2002), Trudgill (2011). CASP captures this as the MCG principle:

(1) Maximize Common Ground (MCG, Filipović 2019: 60)
Bilingual learners and speakers maximize common grammatical and lexical representations and their associated processing mechanisms in their two languages, L1a and L1b.

Filipović & Hawkins (2019) argue that this common ground makes language processing and storage more efficient in the bilingual mind, and it reduces processing effort in the "thinking for speaking" strategies of bilinguals (cf. Slobin 2016). It is also supported by the parallel activation neuro-cognitive model of bilingualism in Costa (2019) and Blanco-Eliorreta & Caramazza (2021). Specifically, Maximize Common Ground makes three general predictions for borrowing and transfer in bilinguals, summarized in (2a–c) (Filipović 2019: 60, Filipović & Hawkins 2019). (L1a and L1b refer to two languages of a bilingual that are roughly equal in proficiency, native-like and acquired simultaneously; if one language is native and the other learned later as an L2 we use L1 and L2.)

(2) a. If L1a and L1b share a given construction, grammatical rule or word meaning, and associated processing mechanisms, then these shared entities will be used more frequently in both languages. These entities may be the preferred or majority pattern in one language and a minority or dispreferred in the other, but they will still be the pattern of choice in the bilingual's use of both languages.

b. If L1a and L1b do not share a given construction, grammatical rule or word meaning, and associated processing mechanisms, then common ground will be created by either introducing entities and rules from one language into the other, or removing them from both. New shared entities will be introduced wherever possible *within* the constraints of current grammatical and usage conventions for the relevant language.

c. Violations of a grammatical or usage convention in L1a or L1b that occur when maximizing common ground (i.e. potential *language change*) will be in proportion to the strength of the environmental and psycholinguistic factors enumerated in Filipović 2019 and Filipović & Hawkins 2019, see Section 3 below.

2.2 Illustrative data supporting Maximize Common Ground and its predictions

With respect to (2a) a minority word order in one language, e.g. adjective before noun (AdjN) coexisting with a majority noun before adjective NAdj, as in French and Spanish, will gain in frequency when speakers of these languages are in bilingual contact with a language that has only AdjN, like English (cf. Nicoladis 2006 for French & English bilinguals, Cuza & Pérez-Tattam 2016 for Spanish & English). Similarly, a pro-drop language in bilingual contact with a non-pro-drop language like English which requires obligatory subjects, will increase the occurrence of its overt subjects (cf. Myers-Scotton 2003 for increased subjects in the Spanish of Spanish & English bilinguals, Schmitt 2000 for Russian & English bilinguals, Savić 1995 for Serbian & English, Polinsky 1995 for Polish, Tamil, Kabardian & English, and Fenyvesi 1994 for Hungarian & English). And a wh-in-situ language like Cantonese in bilingual contact with a language having both wh-fronting and wh-in-situ like English (cf. What did you buy? and You bought what?) will result in more wh-in-situ in English usage by bilinguals (Yip & Matthews 2007).

An example of (2b) involves evidentiality, i.e. the obligatory grammatical expression of the speaker's source of information for the proposition being expressed, whether as direct and witnessed or instead more indirect (Aikhenvald 2004). Turkish is a language with obligatory grammatical marking of evidentiality, and among Turkish and English bilinguals this leads to the more frequent expression of paraphrasing constructions with evidential meaning in English (*it appears that …*, *I am informed that …*, etc), i.e. through a significant increase in the use of optional grammatical and lexical means that are not routinely and obligatorily selected in English but that remain within its grammatical constraints. These optional means maximize the common ground between two languages without actually changing any grammatical conventions in the receiving language (see Slobin 2016 for Turkish & English, Heine & Kuteva 2005 and Aikhenvald 2002 for Tariana & Portuguese).

Examples of (2c), in which grammatical conventions do change when maximizing common ground, include basic word order and even the head ordering typology of languages in bilingual contact with one another. This has been documented across the globe, with productive shifts going both from head-initial to head-final and from head-final to head-initial, reflecting largely sociolinguistic and demographic relations between speakers of the two languages (see Section 3).

The following often quite radical restructurings (even "metatypic" changes, cf. Ross 2007) exemplify changes to one language in bilingual contact with another that *have* taken place in well-defined regions and under sociolinguistic and psycholinguistic conditions that clearly *did* result in language transfers. In (3) we summarize some well-documented cases of VO to OV shifts that were accompanied by cross-categorial shifts to head-final head ordering within other phrasal categories such as adpositional phrases (prepositions yielding to postpositions), noun phrases (noun before adjective and genitive becoming adjective and genitive before noun), etc:

(3) *VO ⟶ OV* in Austronesian languages in coastal New Guinea, e.g. VO *Takia* shifting to OV under the influence of Papuan *Waskia* (Ross 1996);
Yaqui (Southern Uto-Aztecan now with SOV and head-finality) through contact with SOV Hokan and Northern Uto-Aztecan languages (Lindenfeld 1973);
Sri Lanka Portuguese originally an SOV creole, now rigid SOV under Tamil influence (Heine & Kuteva 2005).

In (4) we give examples of the reverse OV to VO shifts, i.e. head-final to head-initial:

(4) *OV ⟶ VO* in Uto-Aztecan *Pipil* through contact and bilingualism with neighboring VO Mayan languages (Campbell 1985);
Papuan *Kuot* (of New Ireland) from Papuan OV to VSO and consistent head-initial orders, surrounded by head-initial Austronesian languages (Lindström 2002);
Eskimo varieties (OV) in bilingual contact with English (Fortescue 1993).

These radical head ordering shifts led to departures from the head ordering conventions in the changing language at the time they were made, and among the bilinguals who first implemented the changes. Some recent studies of Uralic languages in contact with western Indo-European languages provide revealing quantitative evidence of ongoing typological adjustments and of Maximize Common Ground in action. Borise & Kiss (2021) document the gradual emergence of new Russian-type conjunctions and ellipsis possibilities in the Uralic: Finno-Ugric language Khanty over successive 30-year periods beginning in 1900 when intense bilingualism with Russian and Russian dominance began (i.e. their text samples were from 1900, 1930s, 1960s and 1990s). As Russian influence increases over the decades, so these new grammatical conventions emerge and become more frequently used in Khanty. More generally, Kiss (2023) examines the typological cline from OV to VO going from east to west in Uralic languages: the Samoyedic and Ugric subfamilies in Siberia are strictly SOV, the central and northern European Hungarian, Finnish, Estonian and Sami have SVO, and there are flexible SOV-SVO languages in between. In the case of Hungarian, a language with written documents since the 12th century, we see the same OV to VO drift through time. Kiss measures this basic word order drift, and its correlating feature of non-finite subordination marking (OV) versus finite subordination with complementizer words (VO), and links the extent of VO features in space and time to the extent of bilingual influence from surrounding western Indo-European languages. The basic word order alternatives in present-day Udmurt, for example, have been quantified and shown in Asztalos (2018) to be moving more to the Russian VO type among younger bilingual speakers compared with their parents.

This is all very much in accordance with Maximize Common Ground's prediction (2c). The new VO-type grammatical conventions in these erstwhile OV Uralic languages arise under intense bilingualism and in proportion to the extent and duration of the external influence from speakers of VO languages. Where Russian influence is less, for example, in the Siberian Uralic languages, OV features are retained and less common ground is made; where it is greater, in Udmurt, there are increasingly more grammatical changes in the direction of common ground with VO Russian.

3 Constraints on Maximize Common Ground in bilingualism

At the same time Maximize Common Ground does *not* occur, or may occur less, in many bilingual situations that have been observed across the globe, and for a variety of sociolinguistic and psycholinguistic reasons.

3.1 Sociolinguistic constraints

There are many factors of an ultimately social nature that can either limit or enhance the extent to which a bilingual speaker of languages L1a and L1b will introduce convergences between them, through the transfer of linguistic properties from one to the other. So, there is generally less common ground made in formal interactions than in informal ones (Dewaele 2001), and also less common ground when a bilingual is talking to a monolingual rather than to another bilingual, or to two monolinguals in the two languages (Filipović 2019). Maximize Common Ground is also impacted by what Grosjean (2001) calls "language mode", i.e. "the state of activation of the bilingual's language and language processing mechanism". It has been noted that less common ground is made in "code-switching" (Poplack 1980) and "code-mixing" (Muysken 2000) situations, whereby each language retains its essential grammatical properties in the mix and despite the transfer. Perhaps most importantly, there is less common ground made when the feature that might be transferred is found in the socially less prestigious, less dominant and less numerous language (cf. Trudgill 2011 for detailed exemplification of these different social possibilities of relevance for transfer and change in numerous language pairs).

For example, returning to evidentiality, whether a grammaticalized evidentiality morpheme and morpho-syntax are transferred by Maximize Common Ground from an L1a (which has them) into an L1b (which does not) depends on whether L1a is the socially more dominant and prestigious language and on population demographics. Evidentiality *was* transferred from Turkish into Bulgarian, one of the languages of the Ottoman Empire ruled by Turkish speakers (Slobin 2016), but not into higher prestige Greek, despite the same Ottoman rule and hence exposure to Turkish (Lindstedt 2016). Conversely, evidentiality was transferred into socially dominant Andean Spanish from Quechua and Aymara, despite political subjugation, but through sheer strength of numbers of the bilingual speakers using evidentiality in their L1s (Slobin 2016, Filipović 2019). In other words, it is these social variables that crucially determine whether a property transfer will occur at all, and the direction of transfer among the languages in question.

More generally, these social variables either bring about, or limit, the common ground between different languages in these transfers or non-transfers and so exemplify a form of socially sensitive "communication accommodation" between speaker and hearer (cf. Giles & Smith 1979). There is also a growing awareness in the "alignment" literature within the phonetic sciences of the strong social basis for speakers adjusting the fine phonetic details of their pronunciations to their hearers (and even to the voice-AI systems of recent technologies, cf. Zellou, Cohn & Kline 2021 and Zellou, Cohn & Segedin 2021).

3.2 Psycholinguistic constraints

These mainly involve the balanced or unbalanced nature of the bilingualism and comprise both a processing basis and a learning basis. With respect to processing, research on syntactic priming (i.e. the copying of structural preferences among interlocutors when the grammar of the relevant language permits alternatives) has shown that it can occur both within and across the languages of a bilingual (Hartsuiker et al. 2016, Hatzidaki et al. 2011) resulting in a possible preference for the selection of one structure over another (e.g. a passive over an active) across two languages, and hence for a convergence in actual usage between them. The constraints on this convergence, i.e. whether it will actually occur or not, appear to reflect, on the one hand, the degree of structural overlap between the two languages. Passives in German and English do not prime one another, because of the different positions of the passive verb in these two languages (Loebell & Bock 2003), whereas passives in Spanish and English which have similar passive verb positions do prime each other (Hartsuiker et al. 2016). In addition, the more balanced the language command of a bilingual is, the more syntactic priming there appears to be across two languages, and the less balanced the more constrained the cross-linguistic priming will be, especially from the weaker to the stronger language of the bilingual (see Filipović 2019: 41–45 for literature review).

The balanced/unbalanced distinction is also directly relevant for language learning and strongly impacts the degree of convergence. In very unbalanced bilingualism involving L2 learners, common ground between L1 and L2 is regularly created through both positive (i.e. grammatical) and negative (ungrammatical) transfers of L1 features into L2. Definite and indefinite article omission errors by article-less Russian and Japanese learners of English reflect level of proficiency, and these "errors" are then progressively reduced as proficiency improves, resulting in less common ground between L1 and L2 and separate and correct conventions for the two languages (Hawkins & Filipović 2012, Filipović & Hawkins 2013). Depending on the structural properties in question there may be fewer errors and thus less incorrect common ground created between L1 and L2, for example when there are radical differences between two languages as in the ordering of syntactic heads in Japanese versus English. In the study of Cambridge Learner Corpus data by Hawkins & Filipović (2012) there were no recorded instances of Japanese learners converting head-initial phrases of the type [*went* [*to* [*the cinema*]]] into head-final [[[*the cinema*] *to*] *went*] in their L2 English. Similarly, the head-final structures of Japanese are learned early and readily by English learners of L2 Japanese (Rutherford 1983).

The radical head ordering changes summarized in (3) and (4) above and in the Uralic data of Kiss (2023) now look quite remarkable in the light of these L2 data from Japanese and English, which show no errorful common ground. The VO to OV and OV to VO shifts when they occur are explained in Filipović (2019) as a consequence of strong social pressures in favor of convergence between the two languages, and of a form of bilingualism that is both long-standing and widespread (cf. Ross 2007 and Trudgill 2011) and also more balanced, all of which favor the transfer of more complex features from one to the other language of the bilingual. These transfers will initially be errorful but if they proceed gradually and in accordance with ease of innovation (Section 5) they will not impede communication within and across generations and changing grammatical conventions will emerge over time through Maximize Common Ground.

Notice that the negative transfers characteristic of L2 learning, which Trudgill (2011) sees as including many simplifications in morphology and phonology (these are the principal areas he discusses), can also lead to changes in grammatical conventions in the wider speech community that uses the L2, in the event that there are sufficiently large numbers of L2 learners present. For instance, he attributes the extensive inflectional simplifications and levellings in the morphology of mainland Scandinavian languages to the presence of numerous adult Low German speakers north of the Baltic during the protracted period of the Hanseatic League, who would have been bilingual with e.g. Norwegian, but more as an L2 than an L1. A similar argument has been made for many more languages by Bentz & Winter (2013) in their paper 'Languages with more second languages learners tend to lose nominal case', cf. also Lupyan & Dale (2010). For an intriguing proposal quantifying how many L2 speakers are needed to bring about such inflectional simplifications (with particular reference to Afrikaans compared to Dutch), see Kauhanen (2022). For many further details on these kinds of psycholinguistic constraints on Maximize Common Ground and their impact on one or the other language of the bilingual, and possible spread beyond bilinguals to the wider speech community comprising monolinguals as well, see Filipović (2019).

4 General grammatical laws constrain Maximize Common Ground further

4.1 Universal Consistency in bilingualism

Jakobson (1968), Hawkins (1987) and Eckman (2011) have all argued that language learners remain consistent with universal laws, despite their partial acquisition of, and departures from, normal conventions in the ambient language. This is captured in (5) from Hawkins (1987):

(5) Universal Consistency in Acquisition
At each stage in their evolution, first language acquisition stages and interlanguages remain consistent with implicational universals derived from current synchronic evidence.

We propose here extending this to bilingualism in general, as in (6):

(6) Universal Consistency in Bilingualism
Whenever the grammatical conventions of L1a or L1b (or L1 and L2) are changed in bilingualism through *Maximize Common Ground* (2c), these changes are consistent with currently observed cross-linguistic patterns and principles.

In other words, when Maximize Common Ground results in new grammatical conventions in one or the other language (by (2c) in the definition of *MCG* above), these changes will still conform to synchronic patterns seen across languages. These latter will, in effect, constrain and reduce what can be transferred from one language into another at any given time.

Heine & Kuteva (2005, 2006) provide illustrative support for this and for the constraints it imposes on Maximize Common Ground among bilinguals of Sorbian (Slavic) with German and of Molisan (Slavic) with Italian. These bilingual speakers transferred numerous features of German and Italian into their respective Slavic dialects, e.g. the definite article. But not all uses of the definite article were transferred, and not all at once. Importantly, the grammars and polysemy ranges of definite articles in Slavic bilinguals respected the same stages of grammatical shift shown in (7) as is found synchronically in different degrees of definite article usage across the globe (cf. Hawkins 1978, 2004, 2014), from:

(7) demonstrative determiner (*that house*) to
anaphoric definite reference (*a house* ⟶ *the house*) to
associative or bridging definites (*a house* ⟶ *the roof*) to
definite generics (*the lion is a powerful beast, the Italians eat pizza*)

This is the same sequence found over centuries in the attested historical evolution of articles in Germanic and Romance languages (Hawkins 2004, 2014). Bilingual Sorbian does not immediately (or even eventually) acquire all uses of the German definite article: it undergoes the progression in the implicational scale of (7). Some definiteness properties present in German that could have been transferred at successive stages of bilingualism into Sorbian by Maximize Common Ground (2c) were *not* transferred, therefore. The implicational scale structures the sequencing of transfers and changes here.

Moreover, whenever there is an implicational universal (Greenberg 1963, Jakobson 1968, Eckman 2011), or a hierarchy of such implications, there will be a set of constraints on Maximize Common Ground transfers, in accordance with permitted co-occurrences. Consider Keenan & Comrie (1977)'s Accessibility Hierarchy for Relative Clause Formation (SU > DO > IO/OBL > GEN), or any such hierarchy (A > B > C > D). They define a chain of overlapping implications, if D then C, if C then B, and if B then A, and (all and only) the property co-occurrences shown in (8):

(8) A
A + B

A + B + C
A + B + C + D

If the two languages of a bilingual are at different points on this hierarchy, e.g. one has A, the other (A + B + C +) D, then Maximize Common Ground will be attained either by gaining or by losing properties, depending on the sociolinguistic and psycholinguistic factors that determine dominance and direction of transfer, as we have seen. But universal co-occurrences must always be respected in the process and in the interlanguages. The movement in Sorbian and German bilinguals, and in Molisan Slavic and Italian, is from A (demonstrative reference) to B (anaphoric definite) to C (associative definite) and then to some but not all instances of D (generic definites). The sequencing of transfers by Maximize Common Ground is constrained by the definiteness hierarchy: there is no combination of A + B + D without the intervening C.

A further example of universal patterns constraining language change under contact and bilingualism comes from a closer look at some of the word order shifts in (3) and (4) above for which we have historical data. The distribution of basic verb positions and head ordering types in Indo-European ranges the whole gamut from VSO Celtic, SVO Romance/Greek/Slavic, SOVnon-rigid Iranian, to SOVrigid Sanskrit/Hittite (Hawkins 1983). This extreme diversity in one family was argued in Friedrich (1975) to have a language contact explanation. IE word order types correlate with those of neighboring languages in many cases, e.g. Indic SOV mirrors the word orders of neighboring Dravidian languages. Extensive bilingualism and Maximize Common Ground is a plausible explanation for this unusual diversity in IE languages, reflecting their large geographical spread, although further and much more detailed investigation is needed to see whether this drift from OV to VO across Indo-European can indeed be attributed to contact and bilingualism, along the lines of the Kiss (2023) study of Uralic.

Whatever the outcome of this investigation into the causes of typological diversity in Indo-European, it is significant that the language mixtures of early IE were still constrained by general synchronic patterns as found in cross-linguistic samples today. Hawkins (1983) examined the grammars and text counts available at the time for the word orders of early Indo-European languages and argued that they were consistent with (updated) Greenbergian universals, see Hawkins (1979). Specifically, the first attested stages of the IE branches were consistent with Greenberg's cross-categorical word order patterns and preferences (see Hawkins 1980), and text frequencies for evolving patterns through time in selected branches examined were also largely in accordance with the harmonic head ordering patterns described by Greenberg (Hawkins 1983). These cross-categorial word order patterns (VO languages prefer their head categories to be initial or early in their respective phrases, i.e. prepositions initial in PP, adjectives before their complements in AdjP, nouns initial or early in NP, while OV languages prefer the reverse, rigid or non-rigid OV in VP, postpositions in PP, adjectives after their complements in AdjP, and nouns late in the NP, cf. Dryer 1992) have now been explained in terms of Hawkins' (2004, 2014) processing efficiency principle of Minimize Domains. This asserts that the surface strings in which syntactic and semantic relations of combination and dependency are processed should be as minimal as possible, and this can be accomplished, inter alia, when heads of phrases are consistently ordered on one side or the other of their accompanying modifiers or dependents, resulting in the cross-category parallelisms of Greenberg's universals.

These correspondences between Greenberg's cross-linguistic universals and the word order combinations of early Indo-European led Hawkins (1983) to propose the following principle of language change, in accordance with the widely held "uniformitarianism" principle of historical linguistics (the laws of the present are also those of the past, cf. Roberts 2017, and Walkden 2019 for a valuable interdisciplinary perspective):

(9) Universal Consistency in History (Hawkins 1983: 211)
At each stage in their historical evolution, languages remain consistent with implicational patterns derived from current synchronic evidence.

Bilingualism can result in new word orders in languages in contact, therefore, in proportion to the strength of the environmental and psycholinguistic factors we have summarized in Section 3, but these language mixtures are still constrained by Universal Consistency in History (9) and by Universal Consistency in Bilingualism (6) and follow general synchronic patterns found in cross-linguistic samples.

4.2 When changes occur is bilingualism responsible?

We have seen that CASP, in combination with grammatical constraints and the sociolinguistic and psycholinguistic particulars of the contact situation, can clarify when contact does or does not lead to language change. Conversely when changes are observed, CASP can clarify whether contact so constrained is a plausible cause. For example, the regional proximity of Sorbian to German and of Molisan Slavic to Italian, and the socially dominant status of German and Italian respectively, provide compelling evidence for borrowing: these changes did not happen in Serbia or elsewhere in Slavic. The sociolinguistic and psycholinguistic conditions, and the type of bilingualism in question, were favorable to transfer occurring and to Maximize Common Ground. Similarly for the transfer of typological features from Russian into Khanty documented in Borise & Kiss (2021), and from Russian into Udmurt in Kiss (2023) and Asztalos (2018).

Changes in language history may also happen quite independently of contact and bilingualism and in a universally consistent way that responds to the causes and functional pressures that ultimately shape the universals themselves. Seržant & Rafiyenko (2021) give a well-supported example of this in the rise of a head-initial syntactic ordering convention in the history of Greek, resulting in harmonic VO, Preposition before NP, and N before Genitive orders. These were fixed following an earlier word order freedom and mixed typology stage comprising both VO/OV, Prep before NP/NP before Postposition, and N before Gen/Gen before N. They argue that the head-initial ordering convention can be explained through the same functional pressure for head adjacency that underlies the Greenbergian word order correlations themselves (Dryer 1992), as explained now by Minimize Domains (Hawkins 2004, 2014). This new grammatical convention did not emerge in any pre-determined way from the source of the first Greek and earliest Indo-European records, which comprised many alternating orders that could have been, and were, resolved in different branches in favor of either head-initial or head-final conventions (Friedrich 1975). Nor do Seržant & Rafiyenko see or mention any plausible contact or bilingualism scenario that could explain the conventionalization in Greek at the time that it happened. Instead, they see this as a processing-motivated conventionalization of certain word order variants that already existed as options in Greek at the relevant time, whereupon competing orders were eliminated by the kind of gradual conventionalization scenario described in Haspelmath (1999). In other words, they see this as an internally-driven, but processing-motivated, change.

Of course, precisely because changes are universally consistent both when contact and bilingualism are the ultimate cause and when they are not, means that changes can be discussed and analyzed in historical linguistics in purely grammatical terms, and generally have been so, with no reference being made to sociolinguistic or psycholinguistic factors. Add to this the uncertainty about where particular languages were spoken in the past, and the extent and nature of any bilingualism that may have influenced the change, and this neglect was partly understandable. But understanding whether a change was contact-induced can now be considerably helped by having a general bilingualism model such as CASP, in combination with data pertaining to the sociolinguistic and psycholinguistic type of the bilingualism in question. The role of contact in changing the grammatical conventions of Sorbian and Molisan Slavic is now clear (Section 4.1), as is the presence of evidentiality in Bulgarian but not in Greek (Section 3.1), so are the metatypic changes in Austronesian languages of New Guinea such as Takia (Section 2.2), and the ongoing adaptations of Khanty and Udmurt to Russian (Section 2.2). On the other hand, the history of Greek head-initial syntax appears to be a functionally and universally constrained development that is independent of any predisposing source, or of contact and bilingualism. In some cases the ultimate causality of a change may be uncertain, when the genetic affiliations and social demographics and past movements of peoples are less clear, as in the indigenous languages of the Americas (Mithun 1999). We can only hypothesize here that the relevant changes remained universally consistent throughout their histories, whether they were contact-induced or not.

5 Diachronic laws further constrain MCG and universal effects

5.1 Gradualness

The sequencing of changes that takes place when grammatical conventions are altered is constrained by gradualness: it is a fundamental principle of historical linguistics that you cannot change everything at once without jeopardizing communication between generations. Implicational universals and hierarchies constrain the co-occurrences of properties, as we just saw, and define possible (and impossible) sequences of changes. But gradualness then imposes further constraints on the relative timing of these permitted transfers under bilingualism. For example, when L1a and L1b (or L1 and L2) are of quite different word order types, as in (3) and (4), some orders will change before others, in accordance with typological patterns. Shorter modifying categories or dependents, like single-word adjectives, will shift around their head nouns more readily than phrasal modifiers like adjective phrases (Dryer 1992, Hawkins 2014: 85–89, see further below).

Heine & Kuteva's (2005, 2006) Sorbian & German and Molisan & Italian bilingual data involving definite article expansions in these Slavic dialects also exemplify gradualness in addition to universal constraints on language transfers. Bilingual Sorbian speakers did not immediately acquire all uses of the German definite article: the transfers from German by Maximize Common Ground (2c) occurred in a stepwise progression down this hierarchy. Similarly the new conjunction and ellipsis rules of Khanty borrowed from Russian emerged gradually

(Borise & Kiss 2021). Thomason & Kaufman (1988) give numerous examples of gradualness in morpho-syntactic transfers. More extreme departures from earlier systems are eventually found under extensive contact and bilingualism, e.g. the metatypic changes observed between Austronesian and Papuan languages in coastal New Guinea (Ross 2007, Lindström 2002).

5.2 Ease of innovation

There is a further factor that determines the sequencing of any transfers from one language to another under Maximize Common Ground (2c), in addition to implicational universals and gradualness. This involves what we can call the "ease of innovation" for certain linguistic features over others. The basic insight and empirical generalization here is exemplified by degrees of borrowability and the "borrowability hierarchies" of historical linguistics (Weinreich 1953, Moravcsik 1978, Harris & Campbell 1995). For example, lexical items are borrowed more readily than grammatical function words; within the former, nouns more readily than verbs; free-standing (grammatical) words are easier to innovate and borrow than morphological affixes; derivational affixes more readily than inflectional affixes.

We see this exemplified in the early borrowing of free-standing prepositions ahead of words with affixal morphology in the transition from head-final to head-initial word order among Quechua & Spanish bilinguals (Luján et al. 1984). We referred above to single-word adjectives changing their ordering before more phrasal dependents do so, which results in variable orderings across languages such as *yellow book* and *book yellow* in Romance co-existing with consistently post-nominal adjective phrase modifiers like *book yellow with age* (cf. Dryer 1992, recall the discussion of adjective ordering in French & English and Spanish & English bilinguals in Section 2.2).

This mobility for single words is explained by Hawkins' (2004, 2014: 85–89) Minimize Domains principle: single-word changes in word order patterns preserve simple and more minimal domains of processing and are a prerequisite for more extensive and later word order shifts involving phrasal modifiers that require larger and more complex domains of processing when word orders are changed. O'Grady (2020) talks of "Affordability" in this context: if a language permits a more costly operation, it should also permit a less costly operation of the same type. This explains many hierarchies of e.g. filler-gap dependencies and the sequential changes up and down them, including in bilingual transfer, and it explains why less costly changes are implemented first.

Within a generative context certain linguistic properties have been claimed to be easier to transfer into the "common ground" (our term) between two languages if they occur at an "external interface" between syntax and pragmatics (Sorace 2011). For counter-examples and a critique showing that positive evidence and frequency of exposure are actually the more significant factors, see e.g. Özçelik (2017). See also in this context Filipović & Hawkins' (2019) principle of Minimize Learning Effort, which captures interacting factors in CASP that are of relevance for ease of innovation. In more typologically oriented work, Field (2002) proposes that borrowings are easier that are "compatible" with the morphological type of the receiving language (within a morphological complexity scale of: inflectional > agglutinative > isolating).

6 A multi-factor complex adaptive systems approach to bilingualism and language change

We have seen that there are numerous filters on contact-induced language change that collectively serve to limit the properties that *can* be transferred from one language to another at a given point in time, as shown in the funnel in Figure 1.

Figure 1: Filters on contact-induced language change

The properties that make it through to the bottom of the funnel *can* be transferred; those that are progressively filtered *will* not be.

7 Conclusions

Bilingualism has played a significant role in changing the grammatical conventions of one or the other language in the bilingual mind, and these changes have then spread in many cases to monolinguals in the relevant speech communities.

These "bilingualism-induced" changes can accordingly help us answer the "actuation" question raised by Weinreich et al. (1968: 102): "Why do changes in a structural feature take place in a particular language at a given time, but not in other languages with the same feature, or in the same language at other times?"

Our answer is that different types of bilingualism, distinguished according to the many sociolinguistic and psycholinguistic variables in terms of which the types can be classified, help us to understand when certain changes are innovated and in what pairs of languages and speech communities (Section 2), and also crucially when change is not actuated (Section 3). By focusing on bilingualism, rather than contact in general, and specifically by building on an integrated sociolinguistic and psycholinguistic model of bilingualism like CASP (Filipović 2019 and Filipović & Hawkins 2019), we gain greater clarity in understanding change. But even a broadly-based bilingualism model on its own is not sufficient. There are constraints of a universal and grammatical nature on the set of possible bilingualism-induced changes as illustrated in Section 4. Further diachronic constraints involving gradualness and ease of innovation were summarized in Section 5. Hence there are multiple factors that need to be considered of relevance to the actuation problem, as shown in the funnel of Section 6.

Progress has not been made hitherto in understanding when and where changes will be innovated, and when and where they will not be, we believe, because so many discussions of change have not captured the fullness of the variables that need to be considered. They have generally focused on either purely grammatical, or purely sociolinguistic, or occasionally psycholinguistic factors, or on contact rather than bilingualism. CASP already incorporates multiple sociolinguistic, psycholinguistic and grammatical variables, and its predictions for possible versus unlikely or impossible changes are then further filtered by grammatical universals and general diachronic laws. The result is a general model of interacting factors that can be considered when trying to better understand when contact does actually lead to change. Conversely it provides a more precise tool for determining whether changes have, or have not, resulted from contact (Section 4.2).

This multi-factor model can now be applied to contact situations between different languages across the globe and can serve, at the very least, as a checklist encouraging researchers to seek clear data about the bilinguals in question and their balanced/unbalanced status and proficiency levels in the different languages. It will lead them to seek data on the social status, demographics and usage patterns for each language, and it will require grammatical analysis of the typological relationship between the two languages. Universal patterns of co-occurrence will then constrain possible bilingualism-induced changes, along with diachronic considerations of gradualness and ease of innovation. Having multiple factors to consider like this does not make it easy to work out the overall predictions of the model and their expected outcomes. Sometimes different factors co-operate, sometimes they compete, sometimes CAPS' predictions are probabilistic, sometimes they are more discrete. See Filipović & Hawkins (2019) for further discussion and illustration of how different factors interact and work collectively to yield predictions for bilingual outputs, when these can be clearly made. Sometimes the factors pull in different directions and no clear predictions can be made. But very often either absolute or gradient predictions can indeed be made arising out of these multiple factors, as in the kinds of data we have discussed in this paper involving changes in word order (OV to VO, NAdj to AdjN, Section 2.2), avoidance of certain changes (no evidentiality borrowed in Greek, Section 2.2), and sequences of changes with delayed transfer or no ultimate transfer of contrasting features (definite article uses in Sorbian, Section 4.1) The basic point is that we have no choice in this context: language change is determined by multiple factors, so if we are to better understand when contact does lead to change, and conversely when changes have not resulted from contact, we have to consider the full slate of relevant interacting factors, as captured in CASP and the further constraints imposed by implicational universals and diachronic laws. The model presented in this paper defines a broader set of factors than are normally brought together in discussions of language change in the literature, and it is a first step in the direction of making language contact a more precise "cause for hope" in solving the actuation problem.

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Chapter 3

What can toys teach us about actuation?

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This chapter proposes that aspects of the actuation problem can be fruitfully elucidated if the problem is studied in the limited context of simple toy models of variation and change. One such model, the "AB toy", is examined in detail. This examination suggests that the actuation of linguistic change is likely to depend on an interaction of deterministic and stochastic factors. It is pointed out that (simple) mathematical models make available strong probabilistic predictions, and that the solution to the actuation problem, if one exists, is likely to lie in the general system-theoretic properties of speech communities, rather than in the establishment of sufficient conditions for individual instances of change to occur.

1 Actuation, explanation and complex systems

Most discussions of the actuation problem to date have operated under the assumption that if a solution to the problem can be found at all, it must take the form of a deductive-nomological (DN) explanation: a change is actuated if such-and-such conditions are in place, or in other words, the presence of these conditions deductively implies the change (see Walkden 2017 for a recent assessment). Others have pointed to radical differences that are assumed to obtain between the study of social systems (under which language change falls; more on this point later) and the "hard" natural sciences; DN explanation is felt not only to be only possible in the latter and not in the former, but it is also felt that DN explanation fully characterizes explanation in the physical sciences. As a consequence, one is left with the impression that an impermeable qualitative obstacle separates the natural and the human sciences.

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This chapter argues that this view is misguided, and that we can in fact learn much about the actuation of linguistic change by adopting methodological and conceptual tools which are commonly employed in the physical sciences. This, crucially, does not entail adopting DN explanation. Progress in the natural sciences over the last century has shown, time and again, how the nature of explanation itself must change as the object of inquiry increases in complexity. In fact, traditional DN-like laws operating on macroscopic scales, such as those of thermodynamics, emerge from microscopic interactions which can only be understood if the object of interest is viewed as a complex system. Consequently for many everyday phenomena, for instance radioactive decay, only probabilistic explanation is possible. I will argue that social systems – hence also speech communities – are fundamentally similar: they are systems of interacting "particles" (here, individuals with knowledge of language), whose macroscopic properties emerge from their microscopic dynamics. Adopting this stance means that, in many cases, one must forgo the expectation that deterministic predictions can be had in the study of language change. This admission echoes well-known statements of Lass (1980), who has arrived at a somewhat more pessimistic conclusion than I am willing to entertain. I will argue that much can, in fact, be learned about actuation even if our predictions are doomed to be "only" probabilistic.[1]

The second central claim of this contribution is that in order for us to actually learn these things about a complex sociodynamic system, we must be willing to study *simple* representations of reality. Only relatively simple models can be understood, in the sense that they can be given meaningful interpretations and their behaviours studied in systematic detail (in the best case, using analytical mathematical methods). Sometimes the model may be so simple that no one – the modeller included – would pretend that it faithfully characterizes all, or even

[1]Space constraints preclude me from offering a full comparison between Lass's analysis and mine. Briefly, however, the difference is as follows: for Lass (1980), (i) there are no truly DN explanations in (historical) linguistics, (ii) probabilistic explanations are 'non-explanatory' (13) and 'non-empirical' (21) since individual instances can never refute statistical laws, and (iii) there is no other meaningful way in which a probabilistic statement may explain something. I agree with Lass on the first point, and agree with him on the substance (if not the terminology) of the second, but disagree on the third one. This is for two chief reasons. Firstly, I believe it is possible to argue that probabilistic laws about systems do explain observation statements concerning probability distributions, and these have empirical content in the sense that observations made of particular empirical collectives can fit those distributions either better or worse. Secondly, I believe part of the problematic nature of explanation in historical linguistics stems from the reification fallacy which is committed when we take "languages" to be the objects of explanation. In reality, the proper explananda are the collective, emergent properties of speech communities consisting of interacting individuals, and these can only be explained the way the behaviour of ensembles is explained in statistical physics.

most aspects of the object of study. Yet manipulating that model still turns out to be useful. I will call models of this kind *toys*. Just as material toys teach the child about fundamental aspects and properties of the larger-scale objects they represent, toy models can teach us about the more complex phenomena they are models of. Studying these toys leads to a sharper understanding of the *global* aspects of the phenomenon at hand. In particular, I aim to show that consideration of "speech community toys" teaches us something valuable about the interaction of deterministic and stochastic forces in language change. In a sense the solution to the actuation problem (if one wishes to use that term) turns out to be both simpler and more complex than perhaps previously expected – simpler, because the actuation of linguistic change turns out not to differ in any major way from the actuation of changes in complex systems generally, and more complex because the expectation that DN explanation will one day be possible must in all likelihood be abandoned.

Due to space constraints, it is not possible to discuss models in technical mathematical detail in this chapter. This may seem somewhat disappointing, as manipulating toys, interacting with them, figuring out what the presence or absence of this or that part does to the toy's functioning, is an integral part of coming to understand how the toy works. However, references will be given to texts which discuss these important technical aspects in full detail.

2 The AB toy

In the spirit of the above remarks, let us now proceed to examine one of the simplest possible sociolinguistic toys. I will call this the "AB toy", as it describes the competition of two linguistic variants, A and B, in a speech community. The toy assumes the community to consist of N individuals who have the capacity to switch between the two variants. For simplicity, the toy assumes categoricity: at any given time, any given individual is either A or B. The symbolism $X(t)$ will be used to represent the proportion of those speaking A at time t; thus $X(t)$ is a number between 0 and 1 or, more precisely, a number in the one-dimensional grid

$$\left\{0, \frac{1}{N}, \frac{2}{N}, \ldots, \frac{N-1}{N}, 1\right\}. \tag{1}$$

Thus $X(t)$ has $N + 1$ possible values and hops between these values as specified by the system's dynamics, which will be defined next.[2]

[2] Since there are only two variants, $1 - X(t)$ will be the proportion of those speaking B. Hence there is no reason for bookkeeping this second proportion explicitly.

Individuals interact randomly: at random (technically, exponentially arriving) times, two randomly drawn individuals meet, one speaks and the other listens,[3] and the latter may (or may not) update its state depending on that interaction. For simplicity (this assumption will be discussed at length later), we will assume that communication between alikes is always successful: that is, if A and A meet, no update occurs, and similarly for an interaction between B and B. However, whenever the interaction is heterologous (between A and B), the listener may switch their variant. Thus, the following two reactions are both possible (in these reaction schemata, the second summand represents the listener):

$$
\begin{aligned}
\mathrm{A} + \mathrm{B} &\xrightarrow{a} \mathrm{A} + \mathrm{A} \\
\mathrm{B} + \mathrm{A} &\xrightarrow{b} \mathrm{B} + \mathrm{B}
\end{aligned}
\tag{2}
$$

The lowercase letters a and b refer to the rates at which the two reactions occur. They are most straightforwardly interpreted as misparsing probabilities (Yang 2000, Niyogi 2006). Under this interpretation, a encodes the probability of variant B not being able to parse the output of variant A. Consequently, the listener is prompted to switch from using B to using A; the interpretation of b is similar.[4]

Defined this way, the toy is a stochastic process (a continuous-time Markov chain). This aspect of the model has a very important epistemological consequence, namely: it is not possible, in general, to predict (deterministically, in a DN sense) the outcome of the dynamics of the two variants. In the toy model, just as in the real world, we lack detailed knowledge of individual speakers' interaction patterns: there is no way of knowing whether Alice will speak with Bob rather than with Caroline tomorrow.[5] Thus even if linguistic behaviour *within*

[3]The verbs 'speak' and 'listen' here should of course not be interpreted too physically, as if tied to a particular modality. Hence, the two individuals may also sign, or may communicate through writing and reading, and so on.

[4]Although it is practically useful to have a definite interpretation such as misparsing in mind when studying the model, it is important to point out here that this is not the only way parameters such as a and b may arise in a mathematical model of this kind. Alternative interpretations for such parameters include sociolinguistic motivations and functional pressures of various kinds, depending on the application. Similarly, in thinking of the AB toy one need not think of the individuals as switching between the two variants across their lifespans (although this is a natural interpretation in a sociolinguistic setting) – alternatively, each switching event may be interpreted as a failed instance of language transmission across a generational divide, if we simplify by assuming that as soon as the child acquires their language, the parent dies.

[5]Whether this stochasticity is ontological (the world is non-deterministic, for instance because of the existence of free will) or merely epistemological (we do not have sufficient knowledge of all causal chains) is immaterial here.

individuals could be modelled deterministically (which itself is unlikely, but let us assume this for the sake of argument), the speech community will always contain residual *social stochasticity*. From the point of view of explanation and prediction, the upshot is that, in general, only probabilistic claims can be made.

This, however, does not imply that no prediction (or explanation) is possible. Radioactive decay, which was mentioned earlier, may serve as a convenient analogy.[6] It is impossible to predict when any given single atom will undergo decay, much as it is impossible to predict whether any given single speaker will switch variants. However, statements about collectives of atoms can still be derived and expressed, for instance, as half-lives. Crucially, the probabilistic nature of physical law here does not preclude prediction, explanation, or for instance the safe administration of X-rays; it is just that, epistemologically, we need to be prepared to deal with probability distributions instead of point predictions. Something similar is likely to be true of speech communities, too, as will be argued next.

3 Mean dynamic, continuum limit and bifurcations

If social stochasticity arises from the random nature of interactions between individuals, it is natural to wonder what happens as the size of the speech community is varied. Intuitively, if the number of individuals N is increased, the relative influence of any given single individual on the rest of the community must decrease.[7] In the limit $N \to \infty$, individuals become insignificant. The grid (1) becomes the unit interval $[0, 1]$, and the random variable $X(t)$ (the proportion of A speakers at time t) may now assume any value in that interval. This is known as the continuum limit.

It is not difficult to show[8] that in the AB toy, the expected value of $X(t)$, which I will write $x(t) = \mathrm{E}[X(t)]$ for short, satisfies the ordinary differential equation

$$\frac{dx(t)}{dt} = (a - b)x(t)(1 - x(t)). \tag{3}$$

[6] I take this example from Hempel (1966), who discusses radioactive decay as a prototypical instance of probabilistic explanation in the natural sciences.

[7] Strictly speaking, this claim only holds of the sort of setup adopted in the AB toy: random mixing of individuals. In more sophisticated models, in which social network effects are modelled, single individuals may retain large influences even in large populations. These more complex models fall outside the scope of the present chapter, but see, e.g., Fagyal et al. (2010).

[8] The proof consists of formulating the so-called master equation of the stochastic process and finding the first time derivative of the first moment of the probability distribution of $X(t)$. For details, see McKane & Newman (2004), who study essentially the same model but with an ecological interpretation.

This is Verhulst's differential equation, familiar from elementary population dynamics; its solution is the logistic function, familiar to linguists as the prototypical S-curve (e.g. Kroch 1989).

Equation (3), also known as the mean dynamic of the stochastic process (Sandholm 2010), holds for any population size N; it supplies the *expected motion* of the stochastic process. In the continuum limit, however, something remarkable happens: the variation about that expected motion (over separate realizations of the process) vanishes. More technically, it can be shown that every realization of the stochastic process stays within any desired tolerance level from a (deterministic) solution of (3), for any finite time span, in the limit $N \to \infty$.[9] In other words, as population size approaches infinity, social stochasticity vanishes, and we may treat the system as if it was deterministic.

This deterministic limit is very useful for understanding the expected behaviour of the AB toy. From (3), one immediately sees that the system has exactly two equilibria: $x = 0$ and $x = 1$. The former is stable and the latter unstable whenever $a < b$; the former is unstable and the latter stable whenever $a > b$. Thus, whenever stochasticity can be neglected, the speech community will move either in the direction of all-A or all-B, depending on the relative magnitudes of the parameters a and b.

These properties of the AB toy point to one piece of the puzzle presented by the actuation problem. Suppose that, initially, the speech community is all-B, so that $x = 0$. Suppose, moreover, that the two variants are such that $a < b$ (for instance, that A misparses B more often than B misparses A). If a speaker, or a group of speakers now innovates A, the prediction is that this innovation cannot spread: the state $x = 0$ is attracting, and hence the A variant will eventually be lost. If, on the other hand, something conspires to reverse the order of the switching rate parameters, so that now $a > b$, the innovations will eventually become successful. In the theory of dynamical systems, reversals of this kind are known as *bifurcations*. Here, a formerly stable equilibrium becomes unstable, and the speech community is repelled away from it towards the new attractor, $x = 1$.

Why should a bifurcation occur in the first place? Broadly speaking, the reasons may be either language-internal or external. First, interrelationships between different features of language exist, such that a change in one part of the language may trigger a change in another (see e.g. Yang 2000, Heycock & Wallenberg 2013). Secondly, external factors may perturb the stability of equilibria. This may occur due to language contact, such as when the presence of a

[9] This is now often known as Kurtz's Theorem, after Kurtz (1970). The result is not restricted to the AB toy, but applies to a wide class of stochastic jump processes and their expected motions under fairly lenient conditions; see Sandholm (2010) for a rigorous but accessible treatment.

significant fraction of second-language learners in a community sets in motion a change that ultimately propagates through the entire population (Trudgill 2011, Sessarego 2015, Kauhanen 2022). Both processes may also act in tandem, such as when an external factor influences another linguistic feature which then induces a bifurcation in the dynamics of a second feature.

That bifurcations can help to elucidate aspects of the actuation problem is not a new insight: it can be traced back at least to the pioneering work of Niyogi & Berwick (1997); see Niyogi (2006) for pertinent discussion. However, the central role of stability reversals in the behavioural classification of dynamical systems does not seem yet to have made its way into the toolbox of mainstream historical linguists, and so I believe the point bears repeating. As the next section shows, however, bifurcations in (deterministic) dynamical systems cannot be the whole story of actuation: in many cases – in fact quite possibly in all cases of real interest – bifurcations must be squared with finite-population considerations. This, finally, brings us to the point advertised at the chapter's beginning: the interplay of deterministic and stochastic factors in the actuation of linguistic change.

4 Fixation probabilities

If population size is very large, then predictions can generally be based on the continuum limit. In the context of the AB toy, whenever N is large and $a > b$, it is very likely that variant A will prevail in the long run, whereas if $a < b$, it is likely that variant B will prevail. In small populations, however, the stochastic population dynamics interact with the deterministic bias encoded in the rate difference $a - b$. To see this, consider a simple thought experiment. Suppose we have a community of just $N = 100$ individuals, all but one speaking B. Suppose moreover that $a > b$. In this case, the deterministic prediction is that the one innovator, who speaks A, should be able to propagate the innovative variant to the rest of the community. However, there is a non-zero probability that this innovator finds itself in an interaction which prompts it to switch from A to B before it manages to cause another speaker to switch from B to A. At this point, variant A is extinct in the community, so that barring further innovations, the stable equilibrium predicted by the continuum limit ($x = 1$, i.e. all-A) cannot be approached.

Now suppose that, instead of one innovator, we have ten of them.[10] The same

[10] On the empirical question of who innovates and leads linguistic change see Fagyal et al. (2010), Nevalainen et al. (2011), Stevens et al. (2019), Bermúdez-Otero (2020) and Tamminga (2021).

remarks apply: a chain of events that leads to the extinction of A occurs with some non-zero probability.

These observations, while simple, are again epistemologically consequential, for they show that a deterministic bias (such as $a - b > 0$) is in general strictly speaking *insufficient* to actuate a change.

One might wish to throw one's hands up in the air at this point and deem the actuation problem unsolvable – in fact, as suggested in the introduction, this has been the reaction of many. If the successful propagation of a formally advantageous variant is not guaranteed even in a simple model such as the AB toy, what hope do we have of being able to explain the much more complex change phenomena observed in real-life language communities? This reaction is too rash, however, since (i) we can in fact say much more about the preconditions of change in the case of the AB toy, as will be illustrated shortly, and (ii) such facts can even be discovered about considerably more realistic models, albeit with increased technical difficulty (see Blythe & Croft 2021).

Returning to the AB toy, the theory of Markov processes can be utilized to calculate the fixation probability of a mutant A variant in a population of B wild types. Fixation here means, very simply, the event of the population becoming all-A at some time in its evolution (note that this is an absorbing state of the dynamics, since homologous speaker interactions were assumed not to lead to switches), having started from a state of majority-B. For K innovators (in other words, K A speakers and $N - K$ B speakers initially), the probability of the community becoming all-A over time is given by[11]

$$\sum_{i=0}^{K-1} \left(\frac{b}{a}\right)^i \Bigg/ \sum_{i=0}^{N-1} \left(\frac{b}{a}\right)^i. \tag{4}$$

Such results are quite strong, in the sense that they make for exact, quantitative probabilistic predictions. In other words, if we have reason to suspect that a bifurcation has occurred, rendering the B variant less advantageous when in competition with the A variant, and if we can estimate the expected number of innovators in the speech community, the AB toy allows us to derive an exact representation of the probability that the innovations will give rise to a change that goes to completion. This is a particularly strong form of probabilistic explanation: borrowing terminology from Hempel (1966: 67), the "rational credibility of the explanandum", i.e. the amount of confidence we should have in the observation

[11]This follows from the fixation probability of general two-species birth-death processes, which is well-known. For details, see Nowak (2006: chapter 6).

sentence that represents what is to be explained, coincides with the numerical probability of the fixation event in our stochastic model.[12]

5 What constitutes a solution to the actuation problem?

The preceding discussion suggests that actuation ought to be equated with fixation, i.e. the completion of changes. This is not how everyone uses the term: some authors put more emphasis on innovation, essentially identifying the actuation problem with the question of how, why and when linguistic innovations arise. I believe both approaches are needed: to tell a coherent story about actuation, we need coherent stories about both the innovation and subsequent propagation of linguistic variants. The preceding discussion has also suggested that both deterministic and stochastic factors can be expected to play a role in these processes.

It is not entirely clear what *kind* of stories we ought to tell about the component processes, however. It is quite possible, for instance, that many linguistic innovations are similar to genetic mutations in that they are essentially random events (and hence – to pick up our earlier analogy again – as unpredictable as the exact time a single atom will undergo radioactive decay).[13] Change is drawn from synchronic variation (Ohala 1989), and the best story we can tell about that variation may turn out to be probabilistic. Ironically, this crucial aspect is not explicitly modelled by the AB toy, in which we had to "insert" mutations, i.e. innovative speakers, by hand. This was a conscious choice, motivated by the desire not to overshadow the philosophical aspects of the subject matter with technical mathematical details. But in principle nothing prevents one from writing down a mathematical model which incorporates constantly occurring mutations, and in fact several studies have done just that (see e.g. Komarova et al. 2001, Deo 2015, Kauhanen 2020). Whether innovations are modelled explicitly or regarded as external perturbations to a mutationless system, the essential idea remains the same: innovations are random events that are *always* occurring at some probabilistic rate. Whether they end up acting on the speech community, or simply dying away, generally depends on two things: (i) the formal advantage (or lack

[12] For an example of an application of this logic to make and test predictions vis-à-vis empirical data, see the influential paper by Baxter et al. (2009). The authors mathematically derive the expected time to fixation in their model and use this to argue against one particular (pre-mathematical) interpretation of the empirical facts.

[13] Whether mutations are *directed*, i.e. proceed from one variant to another variant preferentially rather than towards any variant uniformly at random, is not at issue here. Many types of linguistic mutation are known to have significant directionality asymmetries (Ohala 1981), but this in itself does not help us in predicting when, or if, said mutations are going to occur.

thereof) of those innovations (as captured for instance in the sign of the rate difference $a - b$ in the AB toy), and (ii) stochastic effects in small (or otherwise highly structured) populations (as captured by the chains of events leading to extinction of innovations in the AB toy).

This brings us to the central question of what, exactly, constitutes a solution to the actuation problem. The original formulation of the problem in Weinreich et al. (1968) lends itself to at least two different interpretations:

1. A solution to the actuation problem consists of being able to specify, in a given historical situation, why a change either occurred or failed to occur.

2. A solution to the actuation problem consists of being able to specify under what conditions a change *would* either occur or fail to occur, possibly through the specification of probabilities of occurrence.

While superficially similar, these statements are very different: the first is particular, and implies that the solution has the force of a DN explanation in actual, specific historical circumstances. The second statement is counterfactual and does not necessitate deterministic prediction.

This difference can again be illustrated using the AB toy. Whenever stochastic factors dominate, such as when both the rate difference $a - b$ and population size N are small in the AB toy, it is not realistic to expect a solution of the first kind. However, a solution of the second kind is always available, in the sense that the model allows us to specify under what conditions change would be observed (or not observed), and with what probability. In other words, whenever a complete picture of a model system's behaviour is available (such as is the case with the AB toy), the unavailability of deterministic prediction in particular cases does not mean that we lack understanding of the phenomenon under study. Understanding the counterfactual behaviour of a faithful model confers understanding of the real-life system modelled, regardless of whether relevant information about the explanans is available in particular historical circumstances.

Naturally, the above should not be read as suggesting that all one needs to do is to set up a mathematical model, chart its behaviours, and compare the output against armchair intuition. Models must always be tested with respect to empirical data.[14] But even here the unavailability of strict DN-style prediction does not

[14]For a (non-exhaustive) selection of modelling studies that take the empirical dimension seriously, see Jäger (2007), Baxter et al. (2009), Deo (2015), Pijpops et al. (2015), Baumann & Ritt (2017), Newberry et al. (2017), Karjus & Ehala (2018), Burridge & Blaxter (2020), Blythe & Croft (2021), Burridge & Blaxter (2021), Kauhanen (2022) and Pijpops (2022). A reviewer points out that, in

strike a decisive blow, but simply means that we must work harder to corroborate or refute models. In short, if our model predicts that change is to be expected with probability p, given that a set of conditions C prevail, what we must do is to study a number of particular historical circumstances which satisfy C, and examine in what proportion of those cases change actually occurred. Should that proportion be close to p, we have at least tentative evidence for the model. This view of explanation admittedly requires further auxiliary assumptions, such as a form of linguistic uniformitarianism – that processes of language acquisition and use are sufficiently similar across time and space. This is not a particularly high price to pay, however, if the alternative is to forgo all hope of arriving at mechanistic explanations of linguistic diachrony.

6 Conclusion

In this chapter, I have suggested that aspects of the actuation problem can be elucidated if the problem is studied in a toy world. Given a sufficiently simple model, fixation probabilities can be calculated for any combination of model parameters; this affords probabilistic explanations of change processes. Needless to say, the model here considered is extremely simple – in fact, it is the simplest conceivable system capable of modelling the replacement of one linguistic variant by another. Slightly more complex models can offer more insight through the

contrast to such studies, the austere type of mathematical modelling defended above runs the risk of becoming "fact-free science" (see de Boer 2012). I agree with this in principle, though with the following proviso: not all empirically relevant research need be empirical, in general, or devoted to the prediction of a particular empirical observation, in particular. For instance, much has been made in linguistics of the occurrence, in much historical data, of S-curves: the S-curve is clearly an explanandum that diachronic linguists should (and do) care about. But it turns out that there exist general properties of complex systems whose presence predicts the occurrence of S-curves, and whose absence predicts their absence, and these properties can be studied theoretically, practically speaking in isolation from further empirical considerations (cf. Blythe & Croft 2012). In other words, even though much science proceeds in the "Aristotelian" way by collecting and systematizing empirical findings, it is sometimes useful to take a Galilean turn and put the theoretical cart before the empirical horse, in order to chart general systems-level properties of the phenomenon under study (see also Feyerabend 1975). The most important reason for doing so is that the abstract study of models, in addition to clarifying the object of study by forcing the researcher to be explicit about definitions, often suggests new *empirical* predictions which a purely empirical approach might not. The AB toy, for instance, as we have seen above, has Verhulst's equation as its mean dynamic and hence predicts an S-curve. However, the individual-based stochastic dynamics now allow us to ask further questions, such as: how long must we wait, on average, to see the S-curve tend to its completion, given a rate difference $a - b$ and a population size N? or how large fluctuations about the expected motion of the dynamics do we expect to see (for instance in corpus data) based on these parameters?

inclusion of further aspects of linguistic dynamics (see again fn. 14). One such aspect was already alluded to above, namely, the inclusion of an explicit mutation parameter in the model. This, together with more advanced conceptions of the advantage enjoyed by linguistic variants, allows more complicated phenomena, such as bistability (the system has more than one stable equilibrium), cycles (the system periodically traverses through recurrent states) and, in higher dimensions, the possibility of deterministic chaos. These aspects of dynamical systems are largely waiting to be studied and made use of by historical linguists.

The actuation of linguistic change may thus be largely equivalent to the actuation of other change processes in other complex systems – in fact, it is a reasonable null hypothesis to think that processes of innovation and actuation, which are at heart population-dynamic, are similar across domains, and only refine this hypothesis if obvious counter-examples are encountered. This suggests the existence of cross-disciplinary synergy benefits. It may therefore be useful to offer a few bibliographical remarks in an effort to stimulate further cross-disciplinary scholarship. The AB toy appears in biological contexts as a non-neutral extension of the Moran process (see Nowak 2006 for an accessible summary, and Clark 2020 for discussion in the context of language change). The toy itself forms a special case of a larger class of models in which the interaction between stochastic population dynamics and the advantage enjoyed by the competing variants may be more complex; such models are studied in evolutionary game theory as stochastic models of imitation (Sandholm 2010). Interestingly, many of these models turn out to have the replicator dynamic (Taylor & Jonker 1978), of which Verhulst's equation (3) is a constant-fitness special case, as their mean dynamic. This is significant from the point of view of linguistic theory, since a number of studies now have offered models of linguistic variation and change using either the classical replicator dynamic or its extension, the replicator–mutator equation, but without specifying how the "phenomenological" equations on the population level follow from underlying, individual-based dynamics (Komarova et al. 2001, Jäger 2007, Deo 2015, Baumann & Ritt 2017, Kauhanen 2020). Further possibilities for modelling the propagation of linguistic innovations are offered by models in which interactions between individuals do not extend across the entire population but are constrained to local environments (Burridge & Blaxter 2021), as well as work that explores the connections between learning theory and population dynamics (Kauhanen 2022).

To conclude, toys can teach us at least the following things about actuation:

1. The actuation of linguistic change will, in many cases, involve a bifurcation in the dynamics of the speech community. Outside of purely neutral

change processes, such a bifurcation signals that the speech community is seeking a new optimum in a complex advantage landscape defined by both language-internal and language-external factors.

2. Bifurcations can be studied in the continuum limit, in which systems become deterministic. However, these analyses must be squared with the fact that actual populations are finite. Finite-system analysis implies probabilities, rather than certainties of occurrence.

3. This means that explanation of change is, at heart, probabilistic. Pure DN explanation should not be expected; but this does not mean forgoing the possibility of explanation or even prediction. Mathematically specified models have the benefit of generating strong predictions (exact probabilities of change/non-change) and should be formulated and studied alongside other forms of theory-building.

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Chapter 4

Parallel linguistic and sociocultural change: Modals and mores

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Grammatical change triggered by sociocultural change has been invoked for English modals of obligation. In particular, the decline of *must* and increase of *need to* has been adduced as evidence of a shift from authoritative to interpersonal obligation. But how can changing functions be tested independently of their grammatical expressions? Here, we do this by assessing the relative frequency and conditioning not only of forms, but also of functions. Drawing on sociolinguistic corpora of Australian English, we establish a replicable three-way meaning classification based on source of obligation. Within a well-defined variable context, *must* is most associated with *Hierarchical* authority, both *have to* and *(have) got to* with *General* circumstances, and *should* and *need to* with *Personal* choice. *Need to* arose rapidly between the 1970s and 2010s, likely as a communal change, favored in 1st and 2nd person contexts and by women and Middle Class men. Expression of Personal obligation also increased, not only with *need to* but across modal forms. Change in the nature of obligation is thus concomitant with, but independent of, changing modal forms. Parallel formal and functional change can be substantiated by applying the heuristic of the linguistic variable to the inherently variable data of everyday language use.

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1 Introduction: Parallel linguistic and sociocultural change?

The actuation problem – "why a change is initiated … at a particular time and place" (Labov 1982: 81) – has been most successfully addressed for externally triggered changes. One example is contact between groups of speakers. Actuation of sound change in contact between speakers of different varieties of the same language has been correlated with demographic data (e.g., Herold 1997), though actuation of grammatical change in language contact has often been merely inferred (Poplack & Levey 2010).

Another candidate external trigger of grammatical change is psychosociological forces or cultural conceptions. The modern development of individualism in the evolution of social relations has been discussed by social scientists, for example, in debates on *Gemeinschaft* vs. *Gesellschaft* societies (e.g. Aldous et al. 1972). A "changing psychology of culture" has also been suggested, and tied to linguistic change, for instance, in the decreasing frequency of words such as *authority* and *obedience* over the last two centuries in U.S. and U.K. books (as observed in Google Ngram Viewer, Greenfield 2013: 1728). Appeals to sociocultural change have been made for changes in the system of English future tense expressions (Nesselhauf 2012: 83) or increasing use of the first-person singular in European journalistic writing (Posio & Heikkilä 2023: 23), but demonstrating a link is still a challenge. Indeed, beyond lexis, the locus of parallel linguistic and sociocultural change has been elusive (e.g., Hilpert 2020). Here, we put forward a step toward the solution, by assessing shifts in both function and form, independently of each other.

To do this, we revisit an acclaimed parallel sociocultural and grammatical change, seen in the English modals of obligation, where change in world view would give rise to a shift from modality functions that are based on social order and general principles to ones that are more "interactive", such as offering advice or making an emotional appeal (Myhill 1995: 160). Notions such as "democratization in discourse" (Fairclough 1992: 201), "colloquialization" of writing (Leech 2013: 108) and "politeness" (Mair 2006: 108) have been extensively invoked to explain changing frequencies of modals. Specifically, a decline in *must* and an increase in *need to* has been linked to a move away from authoritative, towards more personal, obligation.

Language change is gauged by quantitative variation patterns: rising relative frequency of new expressions in variation with existing expressions, in conjunction with shifts in the conditioning of that variation (Labov 1982: 20, Szmrecsanyi

2016, Torres Cacoullos 2012). While variation between competing modal expressions has been examined in several communities, here, we assess not only the relative frequency and conditioning of forms, but also the relative frequency and conditioning of functions, that is, particular meaning nuances, or types of obligation expressed. In this way, we are able to demonstrate shifts in function. This provides a solid basis for probing the relationship between grammatical and sociocultural change.

2 Data

We capitalize on sociolinguistic corpora of Australian English totaling 1.25 million words of spontaneous speech from 259 speakers compiled for the *Sydney Speaks* project (Travis et al. 2023). Recordings were made in two time periods – 1970s-1980s and 2010s-2020s – with speakers of different ages: in the 1970s data, Elderly, born 1900s; Adults, born 1930s; and Teenagers, born 1960s; and in the 2010s data, Adults, born 1960s; and Young Adults, born 1990s.[1] This allows for change to be assessed in real time (between time periods) and apparent time (between age groups within one time period). Participants are stratified according to gender and social class, with three groups established based on occupation, level of education, and high school type – Middle, Lower Middle, and Working Class. Participants all grew up and live in Sydney, and come from the most well represented ethnic groups in Australia, namely Anglo-Celtic, Chinese, Greek and Italian Australians. (More information on the data for this study in Travis 2024, Travis & Torres Cacoullos 2023.)

3 Obligation meanings

The modal meaning of obligation, sometimes called necessity, expresses the existence of conditions, social or physical, that compel the participant to carry out the action indicated by the predicate (cf., Bybee et al. 1994: 177). The literature on English modals of obligation proffers a dizzying array of terms and classifications. Distinctions have been proposed based on a gradation of strength, ranging between strong and weak obligation (Coates 1983: 31–32, 58), and speaker involvement, along a subjective-objective cline (Lyons 1977: 792–793, Palmer 1986: 16).

[1]The three sub-corpora making up the data are identified in examples in the following way: SydS = ANU Corpus of Sydney Speech (Travis et al. 2023); SSDS = Sydney Social Dialect Survey (Horvath 1985); BCNT = NSW Bicentennial Oral History Project (1987). In brackets following each example are the corpus name, the speaker social code – ethnicity, age group, sex – and speaker number; all names given are pseudonyms.

Another proposed distinction considers the source of the compelling conditions as internal versus external to the participant (van der Auwera & Plungian 1998: 80–81). In this view, source constructions from "need" yield participant-internal modal meaning as with English *need to*, those from "have," general participant-external necessity as with *have to*, and ones from "owing" and "duty," deontic necessity as with *should* (van der Auwera & Plungian 1998: 95). The array of classification schemes is indicative of the difficulty of classifying actual occurrences in a way that is replicable and exhaustive.

Close examination of tokens in the spontaneous speech data brings forth a replicable three-way meaning classification, based on source of obligation: *Hierarchical* authority/norms, *General* circumstances/conditions, and *Personal* choice/opinion. This classification corresponds to sources of obligation recognized respectively as: "some person or institution to whose authority [someone] submits; [...] some more or less explicitly formulated body of moral or legal principles; [...] no more than some inner compulsion" (Lyons 1977: 824). It is independent of degree of strength of obligation and of subjectivity understood as speaker authority, but maps onto typologies distinguishing deontic necessity, participant-external and participant-internal modality (van der Auwera & Plungian 1998: 80–81).

Obligation of the Hierarchical authority type is due to an unequal power relationship, typically between parent and child, employer and employee, or teacher and pupil, as in (1), or due to precepts directed at a particular sector of the population that is a minority, or is minoritized, as with decorum for women illustrated in (2). In contrast, obligation related to General circumstances commonly expresses regulations or conventions that standardly apply to everyone at work, at school, as in (3), or in the rules of a game (e.g., *a boy in the middle has to catch*). Also belonging to this second type are situations that are rationally attributed to wider circumstances or that come about as logical consequences, as in (4). Third, obligation of the Personal choice type stems from personal preferences (as in (7) below) or limitations, such as different learning styles (5), and opinions as in (6), where the speaker casts themselves as offering advice rather than as a source of authority.

Implementing "the interpretive component of variationist methodology", as analysts we "infer[red] the meaning or function of each token" (Sankoff 1988: 154), considering the discourse context and drawing on sociocultural familiarity with the speech community. In doing this, virtually all modal tokens occurring in the data were found to be readily ascribed one of these meanings.

(1) Susan: *did you get a punishment for that?*
Ben: *...(1.0) Oh I gotta write lines out.*
[SSDS_GTM_804_Ben, 13:20–13:24]

(2) Gladys: *I had a h- father who was ... very austere in his outlook.*
...(1.5) a --
.. a girl should be a lady,
she shouldn´t be permitted to ... soil her fingers --
...(1.0) for pay,
[BCNT_AEF_079, 01:02:15–01:02:27]

(3) Jo: *we´ve got a set amount of work to do,*
and everybody knows,
they´ve got to get that work done by the end of the period.
[SSDS_ATF_181, 29:44–29:48]

(4) Sophia: *but she has to do it,*
because she has to support her family.
[SydS_AYF_035, 50:31–50:33]

(5) (a teacher describing games she uses in class to help her students learn)
Federica: *some of them have to do,*
.. have [to move] around,
Tania: *[Mhm].*
Federica: *.. in order to learn.*
[SydS_IOF_053, 44:30–44:33]

(6) (re unsolicited advice offered to people suffering depression)
Anna: *it´s not just,*
oh you need to cheer up,
or you need to get it over it,
or you just need some sunshine,
or you should drink more water.
<@ like the @> --
none of that is going to help the neurochemical imbalance,
[SydS_AYF_033, 49:26–49:34]

4 Modals of obligation as a linguistic variable

Modals within the domain of obligation modality constitute a *linguistic variable*, a set of alternating forms (variants) which share a grammatical function (Labov

1969: 728–729, n. 720, Poplack & Torres Cacoullos 2015: 268–270). The quantitative analysis of modals of obligation as a linguistic variable depends on delimiting a *variable context* – the largest environment in which absolute differences between the forms that may exist elsewhere, whether in meaning or distribution, do not come into play (Travis & Torres Cacoullos 2023: 355–363). We first extracted all forms widely recognized as modals of obligation, which revealed five major variants: *must, have to, (have) got to, should, need to* (N = 4,168).[2] We then set aside functions other than the expression of obligation, comprising epistemic modality (N = 240), which indicates that the speaker judges the proposition to be probable and which mostly applies to *must* (N = 190); performative uses (N = 83), which mostly apply to *must* and *should* e.g., *I should say*; and subjunctive uses, which apply only to *should* (N = 32), e.g. *it was a terrible tragedy that he should die.* For a small number of tokens (N = 32), the modality was ambiguous, and these were excluded, leaving a total of 3,781 instances of these five main modals expressing obligation.

Within obligation uses, *have to* is overwhelmingly the majority variant, in large part due to its grammatical flexibility, as, unlike other modals, it occurs robustly in past tense, negative polarity, and interrogative contexts. Thus, the variable context is obligation uses of the five modals in simple Present, positive polarity, declarative clauses. Barely one half of all instances of these modals occurs within this variable context (1,697/3,781). There was a handful (N = 18) of instances that remained unclassifed for obligation meaning type, generally due to truncated or unclear speech, leaving a total of 1,679 tokens for analysis (see Travis & Torres Cacoullos 2023: 360–361 on the coding of the data).

How are the are the five major forms distributed across the three obligation meaning types? Figure 1 depicts these form-meaning correspondences, and shows that obligation meaning types cut across the different modals, justifying their treatment as variants of a single variable. However, it is also the case that the modals have different foci – *must* is most associated with Hierarchical obligation, both *have to* and *(have) got to* with General obligation, and *should* and *need to* with Personal obligation. These meaning associations may be explicable by cross-linguistic patterns of the source constructions (cf. Bybee et al. 1994: 179–180, van der Auwera & Plungian 1998: 81).

[2] *Need to* includes *need*, the two only tokens of which are from the oldest speakers; *(have) got to* includes instances with *have* (contracted and full), and without (*gotta*) (Travis & Torres Cacoullos 2023: 364). Other forms identified were *ought to* (N = 2), *had better* (N = 26) and *be supposed to* (N = 103).

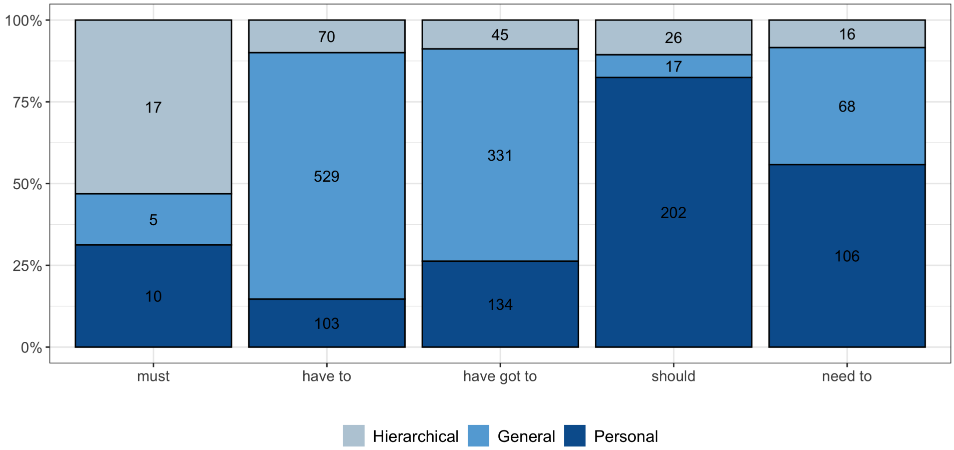

Figure 1: Distribution of obligation meanings according to modal form (N = 1,679)

5 Relative frequencies of forms

For a first view of shifting variation between competing expressions, Figure 2 compares the rate of the different modal forms within the variable context in the two time periods. *Must* was already a minimal variant in the 1970s and is virtually absent in the 2010s. *Have to* and *(have) got to* are the main variants in both time periods, and show a drop in use over time. *Should* and *need to* are the two forms on the rise; for the older form, *should*, this means that it retains its place in the obligation modal system, while incoming *need to*, barely in use in the 1970s, accounts for almost one fifth of all 2010s instances.

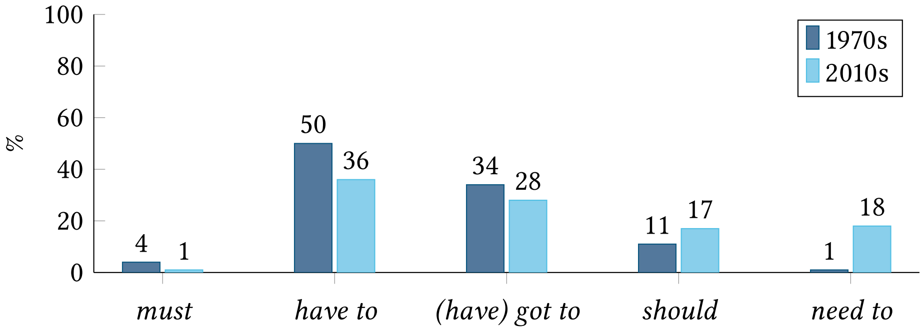

Figure 2: Rate of occurrence of different modals of obligation in the variable context, N = 1,679 (1970s N = 680; 2010s N = 999)

Need to appears to have risen rapidly. It was characterized as "relatively rare" even in early 1990s British English conversation (Biber et al. 1998: 209). Its rate in interviews roughly doubled between the early 1990s and late 2000s, for example, from 6% to 16% of obligation modals in Tyneside, England (Fehringer & Corrigan 2015: 365) and 9% to 16% in Australian English (Penry Williams & Korhonen 2020: 272, 279).

In the present data, there is a steep increase in *need to* in real time, from a rate of 1% (8/680) in the 1970s to 18% (182/999) in the 2010s. In apparent time, comparing within the 2010s, the rate increase is more modest from 13% (57/435) for the Adults to 22% (125/564) for the Young Adults. Not only is the apparent-time difference smaller than the real-time difference, but the 2010s Adults are more similar to the 2010s Young Adults than they are to the 1970s Teenagers (with a negligible rate of 1%, 4/450), who represent the younger versions of themselves (born around the same time, but recorded 40 years earlier). A flat synchronic age distribution that is due to change in both individuals and the community has been interpreted as "communal change", with simultaneous uniform adoption in the community (rather then "generational change") (Labov 1994: 84, Tagliamonte & D'Arcy 2007: 213). This rapid rise of *need to* in the present data suggests that this might be a case of communal change.

6 Conditioning of choices between forms

In order to understand the role of *need to* as a new competing modal of obligation, we consider the conditoning of the variation between *need to* and other modal forms. We fit logistic regression models to the 2010s data (given the negligible use of *need to* in the 1970s), with *need to* as the predicted outcome and obligation meaning type as a predictor, along with subject person-specificity and the social variables of social class, gender and age.[3]

First, we compare *need to* with *have to*, the long-term majority variant (model summary in Table 1). As expected, there is a significant favoring of *need to* by Personal over General obligation, and this exists in interaction with subject, as visualized in Figure 3: *need to* is most favored in Personal obligation contexts with 1st and 2nd person, expressing preference, as in (7), and giving advice, (6) above. There is no effect for any of the social predictors, including for age (with only a small difference between the Adults and the Young Adults, 28% versus 35%).

[3] Mixed-effects logistic regression analyses (lme4 package, lme4 package, Team 2019), with random intercepts for Speaker and Verb (pooling verb types that occur only once, or *hapax legomena*, into a single level).

Table 1: Output of logistic mixed effect model predicting *need to* (vs. *have to*), 2010s

	Est.	SE	*z*	*p*	*N*	*% need to*
(Intercept)	−1.76	0.29	−6.18	**<0.001**		
Meaning (ref: General)					362	22%
Hierarchical	−0.72	0.76	−0.95	=0.35	60	28%
Personal	1.25	0.35	3.63	**<0.001**	196	55%
Subject (ref: 3rd and non-specific)					360	28%
1st and 2nd person	−0.28	−0.36	−0.78	=0.43	258	24%
Subject * Meaning						
3 & Nonspec General					255	23%
3 & Nonspec Hierarchical					18	17%
3 & Nonspec Personal					87	44%
1 & 2 General					107	19%
1 & 2 Hierarchical	1.10	0.93	2.15	**<0.05**	42	33%
1 & 2 Personal	1.43	0.54	2.67	**<0.001**	109	64%

Overall rate of *need to* = 33% (203/618)
122 speakers, variance = 1.13 (SD = 1.06); 109 verb types, variance = 0.95 (SD = 0.97)
Log likelihood: -339.0; AIC: 693.9; BIC: 729.3
(no significant effect for Gender, Social class, Age, Negation)

(7) Sophia: .. *I've never really been like,*
I need to find someone.
I need to get married.
Heather: ... *That's [good].*
Sophia: *[Like that's never been] a priority in my life?*
[SydS_AYF_035, 47:47–47:53]

Second, we compare *need to* with *should*, the two modals most associated with Personal obligation, here considering Personal obligation contexts only (model summary in Table 2). A significant main effect returned in the model is for subject, whereby *need to* is once again distinguished by its co-occurrence with 1st and 2nd person. Co-occurrence with 1st person, recognized as a marker of subjectivity, is consistent with the expression of speaker involvement following from a source meaning of internal need (Aaron & Torres Cacoullos 2005: 616, and references therein). While *need to* and *should* compete as expressions of Personal obligation, this result suggests that speaker considerations are more direct with the former.

There remains no significant effect for age (again, with only a small increase in the proportion of *need to* from the Adults to the Young Adults, 37% versus 45%). However, there is an effect for gender in interaction with social class. As depicted in Figure 4, *need to* is most favored by women and by Middle Class men.

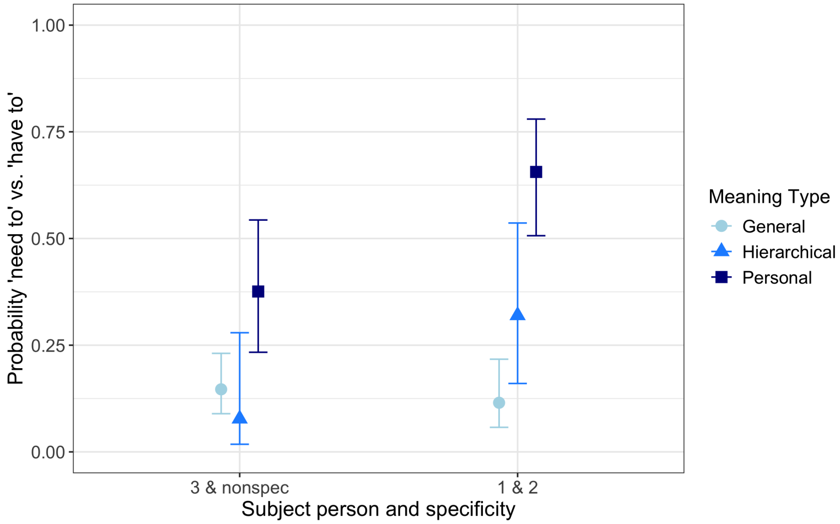

Figure 3: Predicted rate of *need to* (vs. *have to*) for Meaning by Subject (from model in Table 1)

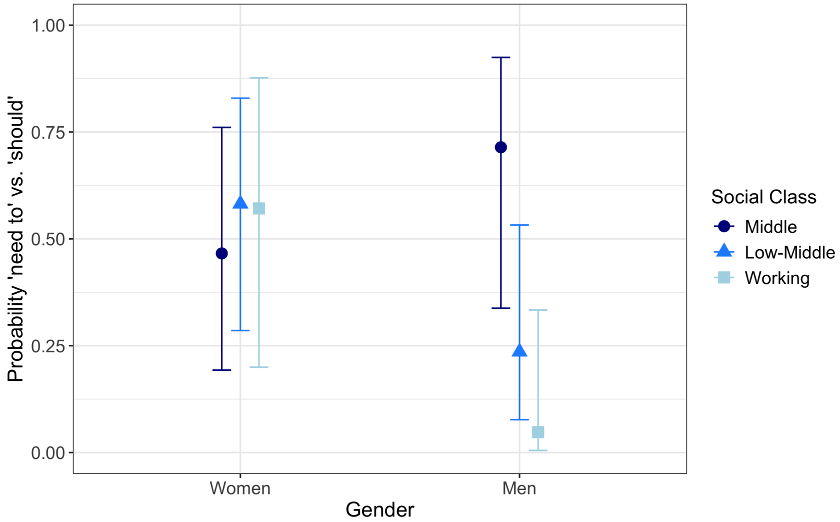

Figure 4: Predicted rate of *need to* (vs. *should*), Personal obligation contexts only, for Social Class by Gender (from model in Table 2)

Table 2: Output of logistic mixed effect model predicting *need to* (vs. *should*), 2010s, Personal obligation meaning only

	Est.	SE	*z*	*p*	*N*	% *need to*
(Intercept)	0.32	0.69	0.46	=0.65		
Subject (ref: 1&2)					142	47%
Nonspecific	−0.43	0.50	−0.86	=0.39	59	44%
3rd person	−2.00	0.66	−3.05	**<0.001**	43	26%
Gender (ref: Women)					139	49%
Men	1.05	1.02	1.03	=0.30	105	34%
Social Class (ref: Middle)					79	44%
Lower Middle	0.47	0.91	0.52	=0.61	111	42%
Working	0.42	1.08	0.39	=0.70	54	41%
Gender*Social Class						
Women Middle					48	40%
Women Lower Middle					60	50%
Women Working					31	61%
Men Middle					31	52%
Men Lower Middle	−2.56	1.43	−1.80	=0.07	51	33%
Men Working	−4.33	1.81	−2.40	**<0.05**	23	13%
Overall rate of *need to* = 43% (104/224); 87 speakers, variance = 3.77 (SD = 1.942)[a]						
Log likelihood: -138.6; AIC: 295.3; BIC: 426.8						
(no significant effect for Age)						

[a]It was not possible to run Verb as a random effect in this model, as out of a total of 62 verb types present, only 18 exhibited variability (36 of which occurred only once in this subset).

7 Shifting obligation meanings

So far, we have seen that obligation modals have preferred, albeit not exclusive, meanings, and that the modal most associated with Hierarchical obligation, *must*, already negligible in the 1970s, has virtually disappeared, while the two that most express Personal obligation – *should* and, especially, *need to* – have grown in relative frequency. Although the decline in *must* and rise in *need to* has been reported in previous studies, what has been lacking is a determination of whether there has been a broader change in the types of obligation people talk about, and whether such a change in function is independent of the rise and fall of individual forms. If the shift in English modals of obligation is associated with sociocultural change, and a move away from authoritative, towards more personal, obligation, then we would expect there to be such a broader shift.

Thus, for our final analysis, we test for a change in obligation meaning over time. Here we examine the entire dataset, and treat obligation meaning as the dependent variable, and predict Personal, in contrast to General and Hierarchical, obligation. Predictors are modal form, together with time period (1970s vs. 2010s), social class and gender, as well as subject person-specificity (model summary in Table 3).[4]

Personal obligation is indeed more likely to be expressed in the 2010s than in the 1970s, with relative frequencies of 42% vs. 20%. It is favored with the modals *need to* and *should* (note that *need to*, with an overall Personal rate of 56%, fails to reach significance due to low token numbers in the 1970s) and with 1st and 2nd person subjects. Most importantly, as depicted in Figure 5, the greater expression of Personal obligation in the 2010s than in the 1970s is not restricted to *need to*, but applies across the modal forms, including the main forms *have to* and *(have) got to*. The exception is *should*, which is already used with this meaning at a very high rate in the 1970s (and thus the significance of the interaction between meaning and modal form in the model is due to the behavior of *should*). (Note, too, that the large error bars for *must* in 2010s and *need to* in 1970s reflect the low token counts, and thus the comparisons via interaction terms for these forms should not be over interpreted).

This across-the-board rise is evidence that the increased talk of Personal obligation is independent of the upsurge of *need to*. Here, then, is a case of "changes in the relative frequencies with which certain functions are expressed" (Myhill 1995: 206). Such meaning frequency changes would be the linguistic correlate of sociocultural change.

As to extralinguistic variables, remember that, compared with *should*, *need to* tends to be favored by women (Table 2 and Figure 4). There may also be a tendency toward greater expression of Personal obligation by women than men (2010s women 46%, 244/536; men 38%, 176/463, $p < 0.05$ by Fisher's exact test). Though not achieving significance in the regression analysis here, such a tendency would indicate that the rise of Personal obligation meaning has been led by women. A leading role for women would be expected from reports of other

[4]Including subject person-specificity deals with possible genre effects on modal frequencies, by factoring in differences between the datasets in the distribution of relevant contextual factors (Travis & Torres Cacoullos 2023: 353–355; cf., Mair 2006: 100).

Table 3: Output of logistic mixed effect model predicting Personal vs. General and Hierarchical meaning

	Est.	SE	*z*	*p*	*N*	% Personal
(Intercept)	−3.49	0.33	−10.58	**<0.001**		
Time Period (ref: 1970s)					680	20%
2010s	1.39	0.35	3.97	**<0.001**	510	42%
Modal (ref: *have to*)						
have got to	1.25	0.36	3.48	**<0.001**	510	26%
must	1.95	0.66	2.98	**<0.01**	32	31%
need to	1.46	1.07	1.37	=0.17	190	56%
should	5.06	0.50	10.20	**<0.001**	245	82%
Subject (ref: 3rd and non-specific)					1151	26%
1st & 2nd person	0.99	0.17	5.91	**<0.001**	528	49%
Time Period * Modal						
1970s *have to*					340	6%
1970s *have got to*					232	18%
1970s *must*					26	27%
1970s *need to*					8	25%
1970s *should*					74	84%
2010s *have to*					362	22%
2010s *have got to*	−0.58	0.43	−1.34	=0.18	278	33%
2010s *must*	−0.84	1.22	−0.69	=0.49	6	50%
2010s *need to*	0.32	1.10	0.29	=0.77	182	57%
2010s *should*	−1.85	0.56	−3.33	**<0.001**	171	82%

Overall rate of Personal Obligation versus General and Hierarchical = 33% (555/1679)
240 speakers,[a] variance = 1.29 (SD = 1.13); 173 verb types, variance = 0.292 (SD = 0.54)
Log likelihood: -754.5; AIC: 1535.1; BIC: 1605.6
(No significant effect for Gender, Social Class)

[a] Of the total of 259 speakers, 19 produced no instances of obligation modals within the variable context.

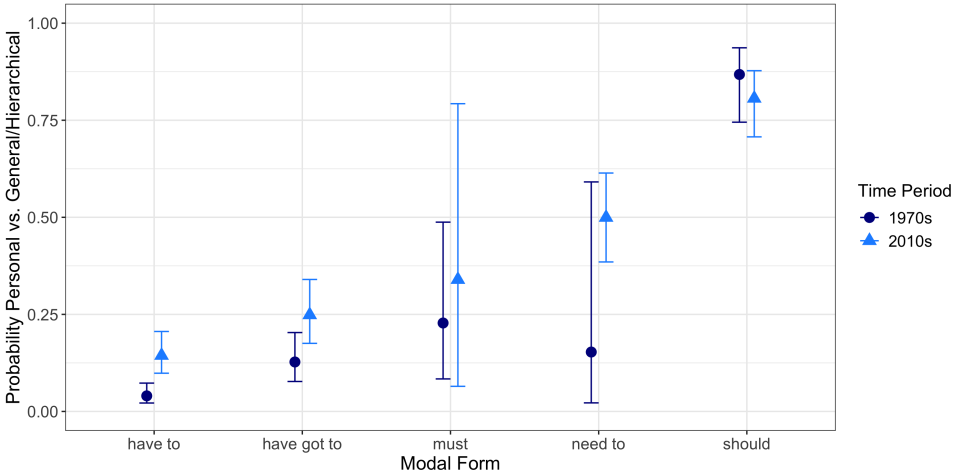

Figure 5: Predicted rate of Personal obligation (vs. General and Hierarchical) for Time Period by Modal form (from model in Table 3)

grammatical changes related to interpersonal relations. For example, a gender asymmetry has been invoked for 18th–20th century changes in Dutch causatives, which have been linked to a decrease in the social role of interpersonal authority (Verhagen 2000: 276). Gender differences have also appeared with the 19th century rise of the English progressive, which has been correlated with the psychosociological factor of interlocutor intimacy (Arnaud 1998: 142).

8 Conclusion

Modals are good candidates for parallel linguistic and sociocultural change, since "… various kinds of obligations … will be culture-dependent …" (Lyons 1977: 825). We expect that material, social change triggers sociocultural innovation, including in language, for example, the Industrial Revolution and French Revolution as the setting for "the Romantic vision" that stimulated the English progressive (Arnaud 1998: 142–144). Nevertheless, as with studies addressing the actuation problem by drawing on facts of demographic change (e.g., Herold 1997), linguists would need knowledge from other disciplines that independently demonstrates a sociocultural change and links it to social changes (cf., Myhill 1995: 201). Here we have addressed parallel change from the perspective of language use, by showing that there have been shifts in function – the relative frequency of obligation meanings expressed – independently of shifts in form – the relative frequency of the modals expressing them. Increased talk of obligation is not solely attributable to the rise of *need to*. In this sense, change in the nature of obligation is concomitant with, but independent of, changing modals.

The actuation problem assumes that the initiation of change is identifiable. However, if language change is generally gradual (e.g., Bybee 2015), what we think of as "innovation" is a posterior recognition of a well-advanced process rather than something we can observe, at least for internal change via grammaticalization (Torres Cacoullos 2012: 108). Thus, actuation is most clearly pertinent for fairly rapid change that is attributable to external triggers. Nevertheless, at the same time, sociocultural triggers for grammatical change may be connected with internal processes such as grammaticalization, by means of the semantics of the rising or receding grammatical variants. Examples are the "warmth" of the English progressive, which suited the Romantic atmosphere of the 19th century (Arnaud 1998: 144); the "tentative" complementizer *fer* 'for' rapidly replacing "confident" *zu* 'to' in the Pennsylvania German of Old Order Mennonites, enacting Anabaptist worldview (Burridge 2015: 48, 51–56); and a steeper decline of "deontic" *must* than other uses of *must* between 1960s and 2000s texts, consonant with democratization (Kranich 2021: 273). For English modals of obligation, variability has been longstanding, with renewal of forms over the centuries via grammaticalization. As grammatical forms have semantic content deriving from their lexical source construction, *need to* readily expresses obligation stemming from personal compulsion (cf., van der Auwera & Plungian 1998: 95). The semantic content of incoming form *need to* would thus accord with sociocultural notions such as individualism.

Morphosyntactic variation and change "emerges through neutralization of distinctions in discourse": in some well-defined contexts, meaning distinctions possibly embodied elsewhere by the alternating forms do not come into play for speakers (Sankoff 1988: 153–155). This hypothesis allows circumscribing the sum of such contexts, the variable context, in order to identify a linguistic variable. Here, within the general function of obligation, we distinguished "specific types of functions" (Myhill 2005: 157): the expression of Personal, General and Hierarchical obligation. Through aggregate quantitative shifts in obligation meanings, we have demonstrated how relationships between grammatical and sociocultural change could be substantiated: the linguistic correlates of changing mores may be sought in the relative frequencies of functions expressed in language use.

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Chapter 5

Sound change in present-day Dutch: A variationist, synchronic approach to the actuation problem

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In this paper we present a variationist, synchronic study of a sound change in progress which aims to contribute to the question of actuation. Previous studies found that Dutch word-initial labiodental fricatives show patterns of devoicing. Completed changes might result in a merger of /v/-/f/. Five regions in the Dutch language area were selected to represent different stages of change. In each region, twenty highly educated young adults participated in a series of production experiments. The sociophonetic analyses of production patterns involve three levels of synchronic comparison: 1) a traditional regional analysis, 2) an analysis of individual patterns within regions and 3) an analysis of the combination of phonetic cues at the individual level. The different levels of synchronic comparison allow us to get a better understanding of the merging process and the individual and group dynamics of sound change.

1 Introduction

Sounds are possibly the most elusive of all linguistic features. Like any other element of the linguistic system, sounds are prone to variability. On top of that, no two realizations (even by the same speaker) are identical due to phonetic and articulatory aspects. This ubiquitous variability sometimes aggregates in a particular direction resulting in a shift in the sound system: a sound change. If sound change is a process intrinsic to language, and if it relies on the variability that is omnipresent in the language, why isn't there change all the time? Conversely,

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why do languages sometimes change, when they could just as well not change? This question has been called the actuation problem and was first raised by Weinreich et al. (1968: 102):

> Why do changes in a structural feature take place in a particular language at a particular time, but not in other languages with the same feature, or in the same language at other times?

The question is thus simple: why does sound change occur? And conversely, why does sound change not occur? In other words, actuation has to explain both change and stability. In this paper, we aim to show how a variationist, synchronic approach to sound change, using the case study of fricative devoicing in Dutch can contribute to our understanding of actuation.

1.1 A variationist approach to sound change

Sociolinguists typically describe natural language variation and its social implications. Variationist sociolinguistics is thus the quantitative study of language variation and change, which establishes relations between linguistic variables and demographic categories, as pioneered by William Labov. It developed in the first place as a reaction against the theoretical study of language, but also to some extent against the traditional focus by historical linguists and dialectologists on the diachronic and spatial aspects of language change. The current paper aims to show how the variationist approach can help to shed a new light on our definition of the actuation question. It attempts to rethink our conceptualization of how sound change proceeds and where it is observable, as it is possibly preventing us from successfully understanding how sound change really actuates.

Firstly, we observe that a traditional two-step view on sound change prevails: sound changes are initiated by a phonetic mechanism, while the spread of these changes is done by social means. Ohala's work in phonetics (Ohala 1981, 1989) first developed the idea that sound changes originate in listeners' misperceptions (e.g., when a listener fails to compensate for coarticulatory effects). The listener fails to attribute the perceived difference to coarticulatory effects and therefore misperceives the sound. It appeared attractive to base sound changes on misperceptions, making their occurrence accidental, and thus non-teleological. This theory also suggests that the key to the actuation problem must lie in the very earliest initiators of sound change. Studying the way in which change subsequently spreads throughout the language community has been left out as a topic for sociolinguists, both in traditional variationist studies (e.g., Labov 1966, 1972, Kerswill

1994, Sankoff 1980) and in later methodological paradigms (e.g., social network analyses in Milroy 1987). It is thus largely assumed that sound change relies on two events that are distinct in nature and occur sequentially: first, the creation of a new variant in speech variation called innovation, and second its propagation, spread in the speech community. This conceptualization in terms of a two-step process of variation and selection draws inspiration from biological evolution (Yu 2013). As pointed out by Labov (1994), however, Weinreich et al. (1968) did not distinguish between these two processes. Indeed, it is quite peculiar to think of sound change relying on two distinct activities: a sort of creative act of deviating from the existing use (of which it is unclear whether it is even observable Croft 2010), followed by the imitation of this innovation by other speakers. In the present study, we intend to step away from the concepts of innovation and spread and reconsider the process of actuation as an iterative process in which minimal changes incrementally accumulate in a speaker's system every time one speaks to a listener or listens to a speaker. In this way, we propose that sound change can be redefined as a purely synchronic process, residing in the variability inherent to production and perception (see also Yu 2023). The propagation of a change, which is perceptible when comparing age groups or diachronic time slices, is then merely a series of incremental acts of actuation in a defined direction. Actuation exists at every step. Change and propagation are the additive effects of such small steps. The current paper aims to illustrate how these incremental acts of actuation take place in a defined direction by observing different phases in the advancement of sound change.

Secondly, we argue that inter-speaker and intra-speaker differences have received little attention in the sociolinguistic and psycholinguistic studies on sound change. In psycholinguistics, inter-speaker differences have frequently been treated as undesirable noise in the data or faded out by pooling or averaging data over speaker groups (Foulkes et al. 2010) with some exceptions (e.g., Abbs 2014, Allen & Herron 2003, Johnson et al. 1993, Mees & Collins 1999). Sociolinguists have also largely made use of methods designed for the community level (Foulkes et al. 2010). The apparent time method for instance (introduced in Labov 1972) has largely been used to investigate change by means of comparing speakers of different age groups. Based on the assumption that our language system becomes relatively 'fixed' after the critical period (Lenneberg 1967), it allows to compare the frequency of a variant in the speech of successive generations. However, one of the major disadvantages of this method is that it can obscure change that takes place within individuals because of life-span change (e.g., Harrington et al. 2000, Sankoff & Blondeau 2007, Trudgill 1988) and age-grading (e.g., Bailey 2002). Some insight into the social characteristics of individual leaders in linguistic change have been provided for instance in the work of Labov (2001) (e.g., leaders in change are typically female, members of the highest status group, upwardly mobile, with dense network connections, with a history of nonconformity in other respects, etc.), but they often remain qualitative observations and are difficult to transpose between linguistic situations and societies. Luckily, sociophonetics and sociolinguistics are now exploring new ways of giving more place to individuals to study variation and change (see also Stevens & Harrington 2014, Tamminga 2021). Does this mean that the key to actuation could be found in the individual speaker? Weinreich et al. (1968) themselves critized theories that situate language change in the individual. They proposed that "a language change begins when one of the many features of speech variation spreads throughout a specific subgroup of the speech community" (Weinreich et al. 1968: 186). While it would be mistaken to think that sound change could directly be found at the individual level, it is nevertheless crucial that we investigate to what extent individual differences can explain sound change. In that sense, the exploration of the individual level could well be the key to understanding sound change. This study intends to demonstrate how to give more weight to individual speakers and their linguistic background when exploring sound change. Thanks to developments in the fields of statistics (e.g., the development of mixed-effects modeling (e.g., Baayen et al. 2008, Barr 2013, Bates et al. 2015, Quené & van den Bergh 2004, 2008), we now have powerful tools to achieve this goal. This paper therefore investigates intra- and inter-speaker variability and tries to bridge the existing gaps between the individual and the group.

1.2 A synchronic approach: Challenges and new perspectives

In the previous section, it was argued that the process of actuation can be conceptualized synchronically as an iterative process in which minimal changes incrementally accumulate in a speaker's system. Researchers interested in the synchronic study of sound change however face a well-known problem: as soon as one finds a sound change in progress, it is already too late to study how it started spreading. It is often assumed that sound change follows a characteristic sigmoidal progression (e.g., Bailey & Haggard 1973, Kroch 1989, Labov 1994, 2001, Chambers 2002). Chambers (2002: 361) distinguishes three phases in the process of the spread of language change: initial stasis, rapid rise and tailing off. Every change starts with a period of initial stability. The exact turning point between initial stasis and acceleration is difficult to determine and appears to be different for every change. As a result, it seems impossible to identify the early phases of change synchronically. It seems that one can only observe new variation patterns and wait until it becomes clear whether they continue along the S-curve into sound change.

We propose an alternative to circumvent this problem based on the observation that change also spreads in the spatial dimension. The spatial aspect is in fact a traditional way to look at how sound change spreads which dates back to the origin of dialectology. Sociolinguists have subsequently concentrated on how change spreads across social classes and age groups. The spatial perspective, however, offers an alternative to the apparent time method (see Section 1.1). It allows for an examination and evaluation of the phases of change in progress and ultimately enables predictions to be made for the change still to come. The observation that linguistic change almost never proceeds in the whole language area at once (except in the case of community-wide change) has given rise to a range of theories describing how language changes across space (e.g., the *wave* model or the *gravity* model, see Trudgill 1974). Change necessarily develops somewhere in the linguistic area (for instance in a politically or economically central area) and spreads across regions because speakers have cross-regional contacts. In that sense, the development of a change through regions within a language area corresponds to different phases of change. The idea of studying diachrony through synchronic is not a novel one, but the regional perspective can specifically help in solving the actuation problem, since change might still need to *actuate* in the population of a certain region, while it is already attested and studied somewhere else.

This idea will be developed in the current study by examining an ongoing merger in the standard Dutch language and its spread across regions. As it will be explained in the next section, we look at a sound change in progress which, depending on the region, is in the initial phase, rapid rise or tailing off stages. In this way, the current study will take this sound change as a case study and get a better understanding of the different phases of sound change by means of synchronic comparisons between individual speakers and between regions.

1.3 A sound change in progress: The devoicing of labiodental fricatives in onset position (/v/ → [f])

Standard Dutch is traditionally described as having a phonological distinction between voiced and voiceless fricatives. The major cue for the voiced/voiceless distinction is the presence or the absence of vocal cord vibration in the fricative (Slis & Cohen 1969). Moreover, voiceless fricatives tend to be longer (Slis & van Heugten 1989) and louder (Kissine et al. 2003) than their voiced counterparts.

During the last decades however, it has been frequently shown in a large number of studies that word-initial voiced fricatives were produced more and more often as voiceless in standard Dutch (Cassier & de Craen 1986, Cohen et al. 1961, Donaldson 1983, Gussenhoven & Bremmer 1983, Gussenhoven 1999, Hamann & Sennema 2005, Kissine et al. 2003, 2005, Mees & Collins 1982, Van de Velde 1996, Van de Velde et al. 1996). The evidence is thus very clear that Dutch fricatives are involved in a sound change, but it seems that not all fricatives are equally advanced in the process. Van de Velde et al. (1996) showed in a real-time study based on radio recordings that devoicing was less advanced in labiodental fricatives than in dorsal fricatives, but further than in alveolar fricatives. In the current study, labiodental fricatives /v/ and /f/ were chosen since they hold an intermediate position within the fricatives undergoing devoicing.

Dutch only has a few minimal pairs with initial /v/ and /f/ (approximately ten) and most of these have a low frequency, e.g., *vee* ('livestock') vs. *fee* ('fairy'). This sound change will very likely result into merger (Pinget et al. 2020). We use the term "merger" to refer to phonological change in which the distinction between two or more phonological categories is collapsed because of a loss of phonetic differentiation (Martinet 1952). Three merger scenarios are possible: in the case of two phonemes, A and B, A and B merge if A moves into the phonetic space of B, if B moves into the phonetic space of A, or if A and B move into a single new phonetic space (Maguire et al. 2013). Labov (1994: 321) distinguished three types of mergers: merger by approximation, merger by transfer, and merger by expansion. The devoicing of labiodental fricatives seems to be a merger by approximation whereby the phonetic targets of two phonemes gradually approximate "until they are non-distinct" (Labov 1994: 321).

Production studies involving standard Dutch have reported significant geographical variation in the development of this merger. Slis & van Heugten (1989) recorded fricatives in the West and the South of the Netherlands (Limburg), and found regional differences. The frequency of initial voicing was lower in the Western region than in the Southern. Van de Velde et al. (1996) showed in their real-time study that fricative devoicing is a change in progress over the entire Dutch language area, but that it is more advanced in the Netherlands than in Flanders. Finally, Kissine et al. (2003, 2005) refined these insights by providing an overview of the differences between four regions in Flanders and four regions in the Netherlands. They found that devoicing of /v/ and /z/ had progressed in Flanders and that the Flemish realizations of the voiced fricatives differed from the Netherlandic ones in duration, noise and pitch. They reported that West-Flanders was the most voiced region and the North of the Netherlands was the most devoiced region. The other regions, Flemish-Brabant, Limburg and Randstad took an in-between position in this order. For duration, they found that the Flemish speakers produced shorter /v/ than the Dutch regions and that the difference in duration between /v/ and /f/ was the largest in West-Flanders and the smallest in the North of the Netherlands. Verhoeven & Hageman (2007) confirmed in an electrolaryngographic study the devoicing of fricatives in Flanders.

In conclusion, there is large consensus and ample evidence that the Dutch voiced labiodental fricative in onset position is undergoing a sound change and is merging with its voiceless counterpart. The progression of this change shows regional differences within the standard variety. The change has been shown to be in the initial or rapid rise phase in most regions and is already clearly tailing off in the Randstad area and especially in the North of the Netherlands. The current study builds upon this large amount of previous evidence on the sound change (/v/ ⟶ /f/) and its characteristics (phonological context and regional spread). The production of this merger is investigated in five regions of the Dutch language area (see Section 2). The study involves three types of synchronic analyses and uses mixed effect regressions to statistically model idiosyncratic variation patterns (see Section 3).

2 Method

A cross-sectional, large-scale production study was designed with a group level consisting of five different regions and an individual level represented by a hundred of participants nested under these five regions (20 in each region).

2.1 Regions

Five regions in the Dutch language area (Flanders and the Netherlands) were chosen to reflect different stages of sound change. Each region belongs to one of the five main dialect groups of Dutch (Flemish, Brabantic, Hollandic, Limburgian and Low Saxon) (Taeldeman & Hinskens 2013). The regions are represented on the map in Figure 1.

West-Flanders (WV) is a peripheral region in the western part of Flanders bordering the North Sea, France and Wallonia (the French speaking part of Belgium). The chosen area is situated in the South of the province West-Flanders around the towns of Kortrijk and Roeselare, and belongs to the Flemish dialect area. This region is known to be the most conservative in terms of fricative devoicing (Kissine et al. 2003, 2005, Verhoeven & Hageman 2007). Flemish-Brabant (BR) is the central area in Flanders and is situated in the economic, cultural, political, and

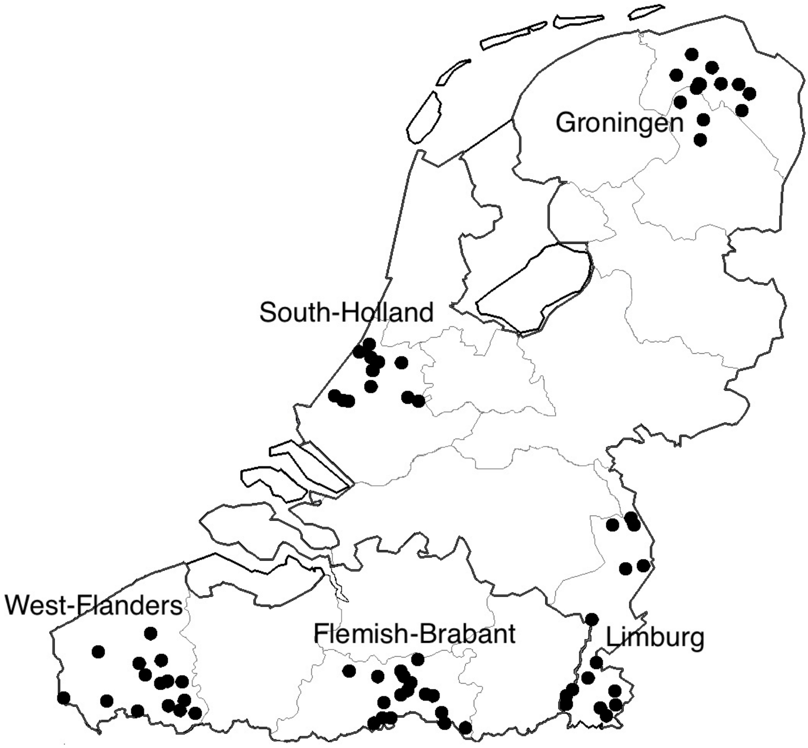

Figure 1: Map of the Dutch language area (The Netherlands and Flanders) and of the five selected regions. Each dot represents the origin of one or more participants.

strongly urbanized heart of present-day Flanders. It belongs to the Brabantic dialect area. Weak patterns of fricative devoicing have been found in this region (Kissine et al. 2003, 2005, Verhoeven & Hageman 2007). Limburg (LI) is a peripheral region situated to the South of the Netherlands. It belongs to the Limburgian dialect area. Participants come from different parts of the Limburg province. The fricative devoicing process is weak to moderate in this region (Kissine et al. 2003, 2005). South-Holland (ZH) lies in the so-called *Randstad*, the central area in the Netherlands consisting of the urban zone in the western provinces of the country. The chosen region centers around the towns of Leiden and Delft, and belongs to the Hollandic dialect region. Production studies cited above have shown strong devoicing of fricatives in this region (Kissine et al. 2003, 2005) and this region is often considered as the area from which this phenomenon is spreading in standard Dutch (Van de Velde et al. 1996). Groningen (GR) is situated in the North of the Netherlands and the participants are recruited from the area around the cities of Groningen and Assen. The region belongs to the Low Saxon dialect area. It is known to be a region where the voiced/voiceless fricative contrast has almost completely faded and resulted in a merger (Kissine et al. 2003, 2005). To sum up, the choice for these five regions built on previous work. Moreover, it is interesting to note that these regional patterns are consistent with the traditional north-south division in the Dutch language area that exists for many patterns in phonetic variation (see e.g., van Heuven & de Velde 2010).

2.2 Participants

The participants were one hundred native speakers of Dutch born and raised in these five regions (see Figure 1). Within each region, 10 males and 10 females participated in the study. The factors *age* and *educational background* were as much homogenous as possible in the participant population. The participants were all highly educated young adults. All of them were attending or recently graduated from a university or a university of applied sciences. They were aged between 18 and 28 years (mean = 22.03) and spoke (colloquial) standard Dutch on a regular basis.

There was no significant difference in age between regions (ANOVA: df = 4; 95, $F = 2.161$, $p = 0.079$). In total, forty participants reported having active knowledge of a local dialect. The proportion of the dialect users differed across regions (df = 4, $\chi^2 = 52.083$, $p < 0.001$) and reflected accurately the current local dialect use situation in Flanders and in the Netherlands (West-Flemish 16/20, Flemish-Brabant 4/20, Limburg 17/20, South-Holland 0/20, Groningen 3/20) (Janssens & Marynissen 2005, Jongenburger & Goeman 2009, Vandekerckhove 2009).

2.3 Procedure and phonetic measures

Participants were recruited at the university campuses of Groningen, Leiden, Nijmegen, Ghent and Brussels. They were recorded in a sound-attenuated booth (in the phonetic lab) of these different universities. Participants conducted five different production tasks: word reading, carrier sentence reading, sentence reading, semi-spontaneous speech, and spontaneous speech. All tasks were designed to elicit the standard variety; they thus did not aim at triggering variables in nonstandard or local dialect varieties. All participants conducted the tasks in the same order.

In the word reading task, each participant read a list of words presented in isolation in a randomized order, including words beginning with labiodental fricatives (n = 20) and fillers (n = 82). In the carrier sentence reading, Dutch non-words beginning with fricatives were produced in the frame "Ik neem de ___"[I take the ___] (n = 18). Next, each participant read a set of declarative sentences in which Dutch words starting with fricatives were elicited (n = 14). Semi-spontaneous productions of fricatives were elicited in two pictures-description tasks. The pictures contained a set of objects that the participants were required to name during the description, containing initial fricatives. Spontaneous speech was elicited in an interview carried out by the experiment leader in which participants spoke about some topics related to their daily life. The interview length was approximately 15 minutes. Ten fricative tokens from the semi-spontaneous production task and five tokens from the semi-spontaneous production task were analyzed, all of them in the onset position of content word, and preceded by a vowel. As a result, each participant produced a maximum of 113 fricatives (57 (v), 56 (f)) (see details of the tasks in Pinget et al. 2020).

All recordings were made with an AKG C420 cardioid condenser head-mounted microphone sampled at 48 kHz (24 bits). All realizations of the target variables were segmented and labeled into phonetic segments within the Praat speech-analysis software package (Boersma & Weenink 2022). Fricatives were segmented by assessing their centre of gravity (Gordon et al. 2002, Jassem 1979, van Son & Pols 1996), following the segmentation protocol for Dutch (van Son 2000) (see details reported inPinget 2015).

The two mainly attested phonetic cues in previous studies (see Section 1.3) were measured on segmented fricative realizations: VOICING and DURATION. Following Kissine et al. (2003), voicing was calculated by measuring fundamental frequency (F0) (in Hertz) with intervals of 10ms in the segment. The presence of voicing was evaluated between a minimum of 50 Hz and a maximum of 400 Hz. The number of measurements with presence of F0 was divided by the total number of measurements and multiplied by 100. This resulted in a relative voicing score, indicating the proportion of each fricative segment as voiced. It is ranging from 0% (no voicing throughout the fricative) to 100% (voicing throughout the entire fricative). Secondly, the total duration of the fricatives was computed in milliseconds (ms). Correlations between the realizations in the different tasks turned out to be high or very high (ranging between r = 0.70 and 0.97), which allowed for their aggregations and made duration normalization unnecessary.

2.4 Analysis

A conservative way of removing outliers was handled. All observations greater than four standard deviations from the mean of the measurements were removed.

In this way, the procedure managed to remove extremely deviant observations, errors in measurements and speech errors ($n = 1039$). In total, 10,261 observations were analyzed (4791 for (f) and 5470 for (v)).

3 Results

The study involves three types of synchronic analyses: 1) a comparison between regions, 2) an analysis of the position of individuals within their region and 3) an analysis of differences between individuals in the implementation of the contrasts. These analyses are presented separately together with an interim discussion.

3.1 Comparison between regions

The first analysis is investigating the region level by comparing the (v) and (f) realizations in terms of voicing and duration. Interestingly, some differences appeared between phonetic realizations of fricatives across different speech styles. Most notably, the word reading task elicited realizations slightly diverging from the realizations in the other tasks. Yet, all realizations were taken together without any further distinction between speech styles for this first analysis, because this study does not aim at investigating style differences and correlations in phonetic measures between styles turned out to be high to very high (ranging between $r = 0.70$ and 0.97).

Results for (v) and (f) split up by regions are presented in Figure 2, a boxplot with voicing measures (in %) in the upper panel and duration measures (in ms) in the lower panel.

Voiced and voiceless fricative realizations showed clear regional differences in both the degree of voicing and duration. The voicing and duration data were fitted with mixed-effects linear regressions using the lme4 package in R (Bates et al. 2015) with the intended voicing category (/v/ or /f/) and region (5) as fixed factor; speakers and words as random intercepts and the intended voicing category by speakers as a random slope. The factors gender and age of the speakers were removed as they did not significantly improve the models. The results of these regressions can be found in Table 1.

Firstly, there was a large, significant difference in voicing between (v) and (f): for West-Flemish participants (the reference level) voiced (v) was produced with much more voicing than voiceless (f) (estimated mean=65.3% vs. 18.0% respectively). These realizations significantly differed from the Flemish-Brabant

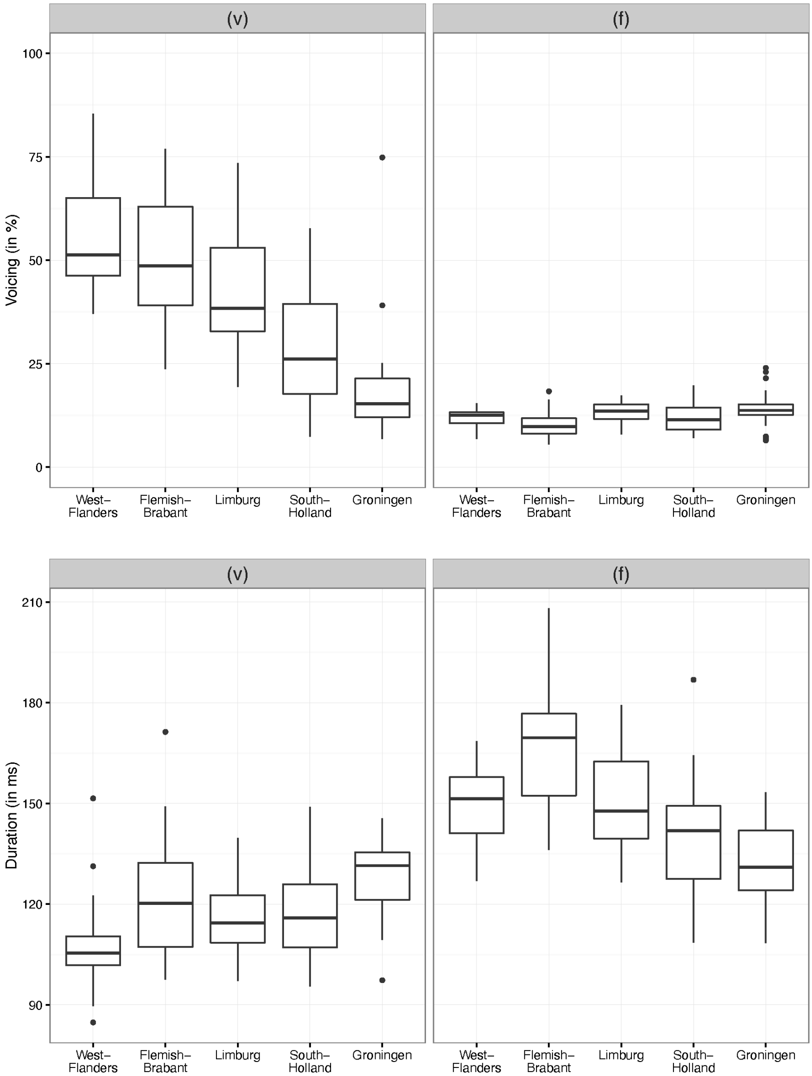

Figure 2: Boxplot of (v) and (f) with voicing (in %) (in the upper panel) and duration (in ms) (in the lower panel), split up by region.

(v) and (f) (estimated mean=59.1% vs. 16.9% respectively). Realizations from Limburg (estimated mean=51.2% vs. 19.1% respectively), South-Holland (estimated mean = 38.8% vs. 18.1% respectively) and Groningen participants (estimated mean=28.8%, vs. 19.7% respectively) all significantly differed from mean voicing in West-Flanders. Furthermore, all regions significantly differed from each other in (v) voicing, but no significant difference in (f) voicing between regions was found, as appeared when changing the reference level.

In addition, there was in all regions a large, significant difference in duration between (v) and (f): duration for (v) was the longest for participants from Groningen (estimated mean=125.8 ms) and the shortest for participants from West-Flanders (estimated mean=107.4 ms) taken as the reference level. In this region, (f) was produced as significantly longer than (v) (estimated mean=154 ms). The durations of (v) produced by participants from Limburg (estimated mean=113.5 ms, $t = 1.32$) and South-Holland (estimated mean=116.3 ms, $t = 1.93$) did not significantly differ from realizations in West-Flanders. The durations of (v) produced by participants from Flemish-Brabant (estimated mean=120.5 ms) did not significantly differ from (v) realizations in Groningen. Overall, the number of hypercorrections (i.e., realizations of (f) as [v]) was very low (less than 1%).

Regional differences found in the amount of devoicing in fricatives matched the patterns described in previous studies (Kissine et al. 2003, Slis & van Heugten 1989, Van de Velde 1996). In all regions, starting with the Netherlandic regions, the contrast in initial labiodental fricatives is fading: in the voiced realizations the amount of voicing is diminishing and the duration of the segment is getting longer, becoming more similar to the voiceless counterparts. The voiced/voiceless labiodental contrast appeared to be almost completely absent in the Northern part of the Netherlands, since the speakers of this region made hardly any distinction in voicing and duration between /v/ and /f/. Caution is taken with this specific region, because in the dialects spoken in that area words starting with voiced fricatives were realized with voiceless fricatives in Middle Dutch dialects (Gussenhoven & Bremmer 1983, van Reenen & Wattel 1992). It is thus possible that initial fricatives in some places have never been realized as voiced, and that this realization from the local dialects is transferred to the standard variety.

3.2 Individual differences within region

This section investigates individual differences in the production of fricatives within each region. In the previous section, it was shown that the (f) realizations were quite stable across regions. In contrast, /v/ is devoicing and merging into /f/. The voicing measures of (v) realizations were therefore fit with a linear mixed effects model. The model intends to control the effects of a range of stylistic and social predictors that are not the target of this individual differences investigation, and which were included as fixed-effects predictors: (1)

Table 1: Results of the mixed-effects linear regressions on voiced and voiceless fricatives (n = 10,261) predicting voicing in % (left) and duration in ms (right).

(a) Random effects

	Voicing		Duration	
	Var	SD	Var	SD
Participant (I)	>0.001	0.009	334.3	18.3
Words (I)	289.9	17.03	188.1	13.7
Variable by participant (S)- (v)	236.7	15.38	5.6	2.4

(b) Fixed effects

	Voicing			Duration		
	Est	SE	*t*	Est	SE	*t*
(I) voiced (v)* WV[a]	65.256	3.519	18.546	107.384	3.269	32.847
Voiceless (f)	−47.282	3.477	−13.600	46.624	2.448	19.043
Flemish-Brabant	−6.200	4.937	−1.256	13.127	4.559	2.880
Limburg	−14.117	4.936	−2.860	6.071	4.558	1.332
South-Holland	−26.515	4.937	−5.371	8.914	4.560	1.955
Groningen	−36.474	4.936	−7.389	18.356	4.559	4.027
Flemish-Brabant * (f)	5.090	4.819	1.056	3.849	3.246	1.186
Limburg * (f)	15.210	4.818	3.157	−6.554	3.243	−2.021
South-Holland * (f)	26.609	4.820	5.521	−19.051	3.249	−5.864
Groningen * (f)	38.182	4.819	7.923	−36.417	3.246	−11.217

[a] West-Flanders

speech tasks (i.e., word reading, carrier sentence reading, sentence reading, semi-spontaneous speech, spontaneous speech), (2) region (West-Flanders, Flemish-Brabant, Limburg, South-Holland, Groningen), (3) speaker gender (male, female) and (4) speaker age (between 18–28 years old at time of interview). The model also includes random intercepts for word and speaker. Following a procedure previously used by e.g., Drager & Hay (2012), Voeten (2021), Tamminga (2021), we took the by-speaker random intercepts as a measure of how innovative or conservative a speaker is in comparison to the other speakers in the dataset within the own region, controlling for all other social and linguistic factors. This procedure provided a determination of every speaker's place on the innovativeness continuum. When random intercepts are negative, speakers are more conservative in the change than their peers. A zero intercept means that the speaker's average voicing production is equal to the regional mean. A positive intercept indicates that the speaker is more innovative than the peers.

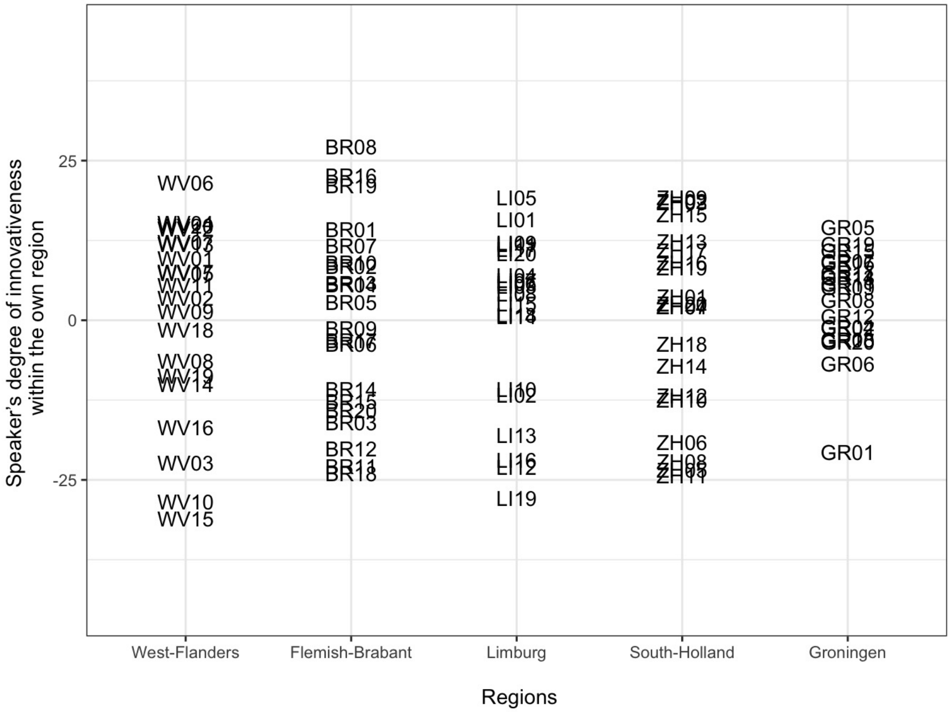

Figure 3: Speaker's degree of innovativeness within the own region split up by region. Individual speakers are labeled with their ID (N = 100).

The degree of innovativeness ranges between approximately -30 and 30. In all regions, we can identify some leaders of change, or at least speakers who are very innovative relative to the relative stage of devoicing in their surroundings, like WV06 in West-Flanders, LI05 and LI01 in Limburg and GR05 in Groningen. In contrast, some speakers clearly appeared to produce /v/ in a more conservative, thus voiced manner, than was average in that region for this population, like LI19 in Limburg and GR01 in Groningen who clearly behaved differently than the rest of the regional sample. Further investigation of additional available information about these individuals (i.e., language attitude, dialect use, personality traits, etc. see details experimental design in Pinget 2015) did not provide further insight into why they specifically behave differently than the rest of their regions.

Furthermore, the range in individual speakers' degree of innovativeness seemed to decrease as the change proceeds across regions: it was larger for West-Flanders and Flemish-Brabant than for South-Holland and Groningen. This observation possibly indicates differences in homogeneity between regions depending on the phases of change. The speakers' population in West-Flanders seemed to be less homogeneous in its production of voicing than the population in Groningen where the sound change is almost complete and where a kind of ceiling effect is therefore reached. This seems to be in line with the idea that the process of sound change actuation is an iterative process in which minimal changes incrementally accumulate in speakers' production system: origin and spread are tied together in each region, rather than two independent events.

3.3 Individual differences within region

As discussed in the introduction (see Section 1.3), many phonetic cues can potentially be used by individual speakers to produce the voicing contrast. We know for instance that – besides voicing and duration measured here – the intensity of the fricatives and the f0 patterns in the following vowel are cues which speakers use to produce the voicing contrast (Pinget & Quené 2023). In this section, we want to examine how individuals combine phonetic dimensions to produce fricative contrasts and to what extent sound change can be described as a shift in the use of phonetic uses.

The use of the voicing and duration per individual speaker is presented in Figure 4. In this graph, voicing is plotted along the x-axis and duration along the y-axis. The mean voiced realization is represented – for each participant – by a filled circle and the mean voiceless realization by a cross symbol. A solid line was drawn from the voiced to the voiceless mean measure. The length and the inclination of the lines give insight into the individual patterns of phonetic cues. The longer the line, the stronger the individual's contrast between the two categories. A more horizontal line means that the participant mostly used voicing to make the contrast. A more vertical line means that the participant mostly used the duration cue. A diagonal line shows that the participant used both voicing and duration to produce the voiced-voiceless contrast.

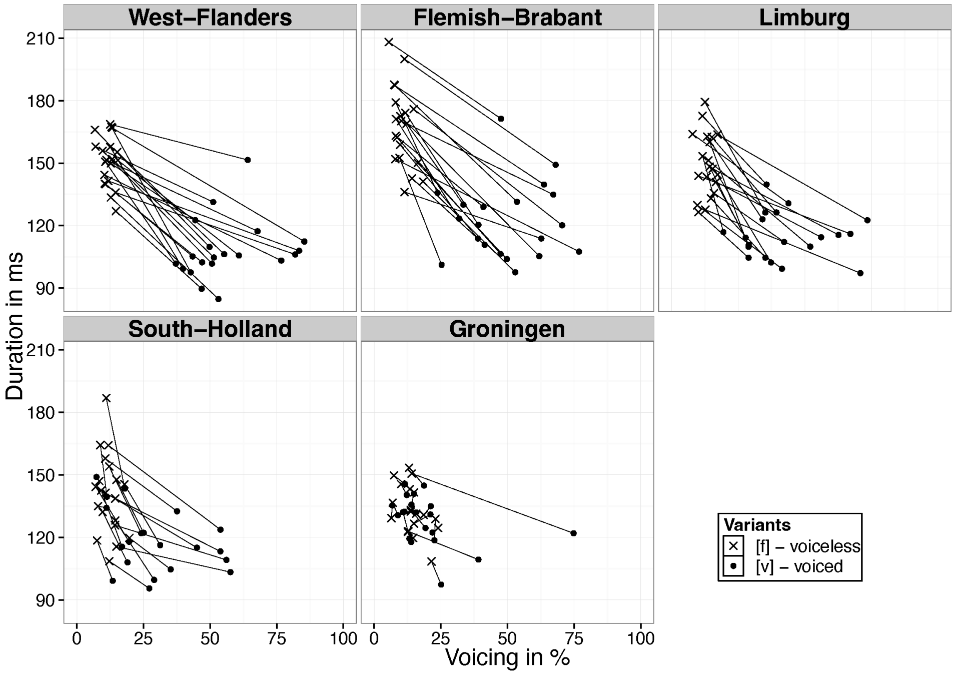

Figure 4: Combined use of the cues in fricatives production split up by individual and grouped by region: voicing (in %) on the x-axis and duration (in ms) on the y-axis. Mean target voiced fricatives are represented by filled circles and target voiceless fricatives by crosses ($N = 100$).

(f) realizations show – as expected in all five regions – a high degree of homogeneity, whereas average (v) realizations show a lot of variation. In West-Flanders, Flemish-Brabant and Limburg, productions are characterized by diagonal lines, meaning that all used both voicing and duration in the fricative contrast. The lines of Limburgian participants tended to be shorter, indicating the contrast is fading due to a reduced use of both phonetic cues simultaneously. Interestingly, the pattern for South-Holland showed that around half of the participants had relatively short diagonal lines. The other half showed very short and sometimes almost vertical lines, which illustrates the loss of the contrast or – when the contrast is still to some extent maintained – the use of duration solely to produce the contrast. Finally, in Groningen most participants had no line at all (i.e., merger), or a very short vertical line (a small difference in duration). Only a small number of participants stood out as using both dimensions to a much larger extent than their regional peers. In conclusion, the analysis of two different phonetic cues in fricatives showed a clear gradual devoicing pattern at the individual level. The process of devoicing starts with a diminished reliance on voicing to produce the contrast. In a second step, it is accompanied by a diminution of the duration contrast. These tendencies are visible in each community which reinforce the idea that such a gradual sound change can be seen as a series of incremental acts of actuation in a defined direction. In other words, actuation seems to exist at every step: change and propagation are in each region the additive effects of such small steps.

4 Discussion and conclusion

In this study, the production of a consonantal merger was investigated in five regions of the Dutch language area. The study presented three types of synchronic analyses: 1) a traditional comparison between regions, 2) an analysis of the position of individuals within their region, and 3) an analysis of the different cues used by individual speakers to implement the contrasts that is fading. Moreover, attention was paid in statistical analyses to the modeling of idiosyncratic variation patterns by means of mixed-effect modeling.

Dutch labiodental fricatives appeared to be in an advanced phase of merger, as expected: rapid rise and tailing off, following Chambers' terminology (2002: 361). Regional differences found in the amount of devoicing in fricatives matched the patterns described in previous studies (Kissine et al. 2003, 2005, Slis & van Heugten 1989, Van de Velde 1996). Logically, the sound change appears to be further advanced than reported two decades ago by Van de Velde (1996) and Kissine et al. (2003, 2005). Both regional and individual differences were shown in the use of voicing and duration to produce the contrast, with in the most advanced region a full merger for almost all individuals between the voiced and voiceless category.

The (range of) measurements along the voicing dimension allow us to precisely define the mechanism of category merging in production. Before the change, the situation was stable with two separated categories of comparable range (A and B). At the change onset, the changing category (A – in both cases the voiced one) increases its range, while the difference between the categories decreases in a linear manner. Further in the change, the range of the changing category starts to decrease, while the difference between categories keeps shrinking. At the change completion, both categories are merged (no or very small difference between categories) and have a comparably small range.

All three analyses made use of mixed effects regressions allowing the differences between individuals (and between linguistic items) to be accounted for in the random part of the model. This attention to idiosyncratic variation patterns appears to be crucial to understanding the different phases of change. For instance, the degree of individual speaker innovativeness obtained in the second analysis provides insight into the homogeneity of the speaker's population. This degree of innovativeness obtained by taking the by-speaker random intercepts in a statistical model controlling for all available social and linguistic factors is a very promising procedure. In future studies, the degree of innovativeness could be correlated to any social, personal or cognitive factors which have been claimed to potentially play a role in language change, such as personality traits, mobility, short- and long-term memory etc. Pinget (2022) makes a first step in this direction, presenting a study of a possible correlation between the individual degree of innovativeness in sound change with imitation capacities. Other scholars are now venturing along similar lines and experimenting with agent-based modeling to investigate the role of individual phonetic variation (e.g., Stevens & Harrington 2022).

Even though the current study provided a much larger number of speakers than usual in this type of sociophonetic work, the question remains open whether twenty speakers could be representative for a region and whether there exist different speaker subpopulations in each region during sound change. Future work based both on the analysis within regions and the analysis of the combination of phonetic cue might help tackling further this issue. Moreover, this type of sociophonetic work on speech production can also further be enriched by the investigation of other aspects of the linguistic competence, like speech perception, imitation capacities, and data on linguistic attitudes and salience. To sum up, the study of language variation and change could largely benefit from more synchronic or diachronic studies involving data from different linguistic aspects, but essentially gathered for the very same individual participants, as only this kind of experimental designs will make the abovementioned types of analyses possible to conduct (see also Pinget et al. 2020).

The ongoing devoicing of labiodental fricatives in Dutch was used as a case study here. The gradual character of this sound change in progress was at the heart of this study. Sociophonetic research has largely been skewed towards English vowels (Foulkes & Docherty 2006). Vowels are typically seen as carriers of sociolinguistic variation, and consequently too little attention has been paid to consonants. Moreover, consonantal variables have often been analyzed auditorily rather than acoustically (Foulkes et al. 2010). The presented study is tackling these issues, but the question might arise to what extent the reported patterns are typical for phonetically "gradual" changes as opposed to "discrete" changes. Labov (1981) indeed argued that sound change may be implemented either discretely or gradually. We observed however that even so-called 'discrete' changes proceed somehow gradually (e.g., by means of lexical diffusion), so that change is always characterized by an incremental probabilistic shift of some sort. Phonetic changes, including the one studied here do not proceed in large, discrete steps, but rather along continuous scatters of variability in individual systems. It is the researchers' task to study the change and the complexity of its graduality looking at the relevant factors. Those might differ depending on the change under consideration.

Finally, one of the crucial insights is that we do not need to (desperately) strive to find (sound) changes in the earliest phase of change to solve the actuation problem. There are ways of circumventing the problem of not being able to distinguish between variation and change. Since the introduction of the apparent time method, it is clear that synchrony can offer a window to diachrony by means of age group comparisons. The alternative proposed in this study in the same synchronic line of thinking was to compare populations along the spatial dimension. All variation and change can ultimately be viewed as the outcome of some form of contact between different individuals or members of different groups. Future multidisciplinary research is required to understand all aspects of present-day regional linguistic contact in all its complexity (e.g., data on mobility, influence of spoken and written media, etc.). The present study already demonstrated how investigating sound changes at the regional level corresponds to examining different degrees of change completion. In this way, it provides a way of observing the iterative process of actuation at each step of completion. The propagation of a gradual change is a series of incremental acts of actuation in a defined direction. Change and propagation are the additive effects of such small steps and are observable at different phases in the advancement of sound change. It is of course an open question to what extent this can be extended to other cases of sound changes with uneven or more abrupt propagation patterns or/and change reversals.

In conclusion, the approach adopted in this synchronic study shows that we can advance our understanding of actuation by improving shortcomings in the study of sound change which are related to the way actuation is defined, by studying sound change at both the individual and group level and using the necessary empirical tools to explore change. The specific insights provided by the different levels of analyses (regional, inter- and intra-individuals) highlights the importance of including them in further sociophonetic analyses in order to achieve a better understanding of the relationship between individual and group dynamics in sound change in progress.

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Chapter 6

Relative clause-forming strategies in German: On typological generalizations and diachronic language change

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In this article we explore the question of whether synchronic typological generalizations such as the Noun Phrase Accessibility Hierarchy (AH) (Keenan & Comrie 1977) can help to understand diachronic language change in a micro-typological approach. We compare relative clause (RC)-forming strategies in Upper German dialects (Bavarian and Alemannic) and written standard German with those in the historical dialects of earlier time stages (1350–1800). We suggest that language change, as it has occurred in German in terms of its RC-forming strategies ([+/− case RC strategy]), is to some extent predicted by the AH or continuous segment prediction. Very specific changes in relation to the form of the relative marker, on the other hand, cannot be explained by the AH.

1 Introduction

The most basic question in the context of variation and change is that of the causes of variation and change: Why does a new form suddenly appear, in a system that has (apparently) worked without problems up to now, and for a function for which one (or even more) forms already exist? If this structure or form persists (i.e., does not remain a slip of the tongue, a transcription error, or a one-off event) and is integrated into the language system, then we speak of language change. We could then also go a step further and situate the above question in a (micro-)comparative context: "Why do changes in a structural feature take place

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in a particular language at a particular time, but not in other languages with the same feature, or in the same language at other times?" (Weinreich et al. 1968: 102). The latter question, the so-called "actuation problem", will very probably continue to remain unanswered, but our study is nevertheless intended as a step towards a better understanding of the question: why is this change occurring here and now?

The aim of this paper is to examine whether synchronic typological generalizations such as the Noun Phrase Accessibility Hierarchy (AH) (Keenan & Comrie 1977) can help to understand diachronic language change in a micro-typological approach. We will compare relative clause (RC)-forming strategies in Upper German dialects (Bavarian and Alemannic) and written standard German[1] with those in the historical dialects of earlier time stages: While in the historical dialects (1350–1800) relativization is possible in subject (SU) and direct object (DO) position with two different RC-forming strategies at the same time ([+ case and – case RC strategy]) in the modern dialects (including the written standard language) SU and DO position can only be relativized with either a [+ case RC strategy] or a [– case RC strategy]. We will argue that this change in strategy has to do with the continuous segment prediction as exemplified by Keenan & Comrie (1977) in their cross-linguistic data. Changes in relation to the form of the relative marker, on the other hand, cannot be explained by the AH. Instead, language internal mechanisms and language-external processes must be considered.

The article is structured as follows: We begin with an introduction into the linguistic geography of German and into RC formation in German (dialects). We then discuss the extent to which the historical dialects can be regarded as precursors of the modern dialects (1800-); after that we shortly present our data and methodology (Section 2). In Section 3, we introduce the AH (hierarchy constraints, syntactic positions, RC-forming strategies) and address the question of whether and in what form it can be applied to German (historical) dialects. In Section 4, we look at RC-forming strategies in German (historical) dialects and discuss the extent to which synchronic typological generalizations can make predictions about (diachronic) language change. Finally, we turn to changes on the form side in the individual dialects which the AH cannot explain (Section 5).

[1]By "written standard (language)" we understand the language, which is codified, taught formally and which has official status in the educational system (cf. Ammon 2005: 32).

2 German (historical) dialects

2.1 Linguistic geography of German

Roughly speaking, the German language area can be divided into three large dialect areas: Upper German (with Alemannic and Bavarian), Central German (with West and East Central German), and Low German (with West and East Low German). This classification is mainly based on phonological distinctions, as we find them in written dialects in the 19th and early 20th century. When we talk about modern dialects (in this article), we are referring to a period from 1800 onwards. We refer to the period before that (1350–1800) as "historical dialects" (this includes the period "Early New High German" from 1350 to 1650 and "early Modern (New High) German" from 1650–1800). The following figure is based on the maps by Wiesinger (German dialects before 1945, cf. Wiesinger 1983: 31) and Reichmann (Early New High German *Schreiblandschaften* 'writing landscapes', cf. Reichmann 1986: 119). The two maps are, in terms of their linguistic and geographic classification, identical since the spatial boundaries of the historical dialects have not significantly changed from those of the modern dialects (cf. Reichmann 1986: 119).

Figure 1 layers the relevant information from both Wiesinger and Reichmann: at first glance, the historical dialects with its different writing/printing landscapes can be recognized. A closer look reveals that the modern dialects can be found within them/are identical to them. The historical dialects are usually divided into 4 or 5 dialect areas (cf., e.g. Hartweg & Wegera 2005: 30, Reichmann 2000: 1625): East Upper German, West Upper German, East Central German, West Central German. The term "North German" (Reichmann 2000: 1625) describes (written) High German in the Low German dialect area. This is because, from the late 15th to 17th century, High German gradually replaced Middle Low German in the whole public written domain (e.g., books and official documents were now printed in High German, education at school took place in High German) (cf. Langer 2003: 291–293). The Low German language area played no role in the development of a written standard language.[2]

[2]One of the main reasons for the loss of Middle Low German (and its failure in terms of standardization and codification, cf. Langer 2003: 298) is the decline of the Hanseatic league, and with it the loss of importance of the northern German area and the rise of the more southerly regions (cf. Langer 2003: 293). By 1650, the shift towards High German as written standard variety on Low German ground is complete (cf. Langer 2003: 291). The period after 1650 is called "Modern Low German" (cf. Langer 2003: 284).

Figure 1: German (historical) dialects

2.2 Relative markers in German

Relative markers in modern German varieties are postnominal, introduce a relative clause, and appear either in the form of a resumptive, inflecting relative pronoun (*d*-pronoun or *wh*-pronoun) or an invariant relative particle (*wo*). We are in particular interested in RC-forming strategies with *wo* as this particle represents the RC-forming strategy in Alemannic, is part of the Bavarian strategy, and is used as locative and temporal relative in written standard German (and the dialects). In the following example *wo* relativizes the subject position of the relative clause; to illustrate and introduce the terms, the relative clause is underlined, the **relative marker** is bold, and the head noun of the matrix sentence is double underlined:

(1) Alemannic (Swiss German)

ich	habe	einen	götti	gehabt	**wo**	gut	situiert	gewesen ist
I	have	an	uncle	had	where	well-to-do	been	is

'I had an uncle who was well-to-do'

Scherrer et al. (N.d.: speaker ID HEll1063)

In contrast to Alemannic, *wo* is used together with a *d*-pronoun in Bavarian. The following example shows *d*-pronoun + *wo*, relativizing the subject position of the relative clause:[3]

(2) Central Bavarian

Das	ist	der	Mo,	**der**	**wo**	uns	g'hoifa	hod
This	is	the	man	who	where	us	helped	has

'This is the man who helped us'
After Pittner (1996: 135)

While in written standard German, subject and object positions of the relative clause are usually relativized with a *d*-pronoun (rarely with a *wh*-pronoun, see Section 5.3), *wo* can be used as locative (and temporal) relative: *Wo* can relativize noun phrases which are in a locative (or temporal) relation to the verb of the relative clause (cf. Zifonun et al. 1997: 42). In the following example *wo* relativizes an NP with locative semantics:

(3) Written standard German

Das	ist	die	Stadt,	**wo**	ich	wohne
This	is	the	town	where	I	live

'This is the town where I live'
After Eisenberg (2020: 300)

2.3 Written historical dialects

We would like to know whether these differences in form and function were already attested in the written historical dialects.[4] As mentioned in Section 2.1, Alemannic corresponds to the former West Upper German and Bavarian to the former East Upper German area. The West Upper German area, with its main writing/printing centers Basel, Zurich, and Strasbourg (see Figure 1), maintained its dialectal character even in the written variety as it was not involved in standardization and codification processes since 1500 (cf. Mattheier 2003: 216). East Upper German, with the writing/printing centers Augsburg, Nuremberg, and Vienna (see Figure 1), was influenced by diffusion and levelling processes: "[One] could have expected the formation and diffusion of a German standard language

[3] In some Bavarian dialects the relative particle can be *was* 'what' instead of *wo* 'where' (Fleischer 2004: cf.).

[4] It should be noted that the written historical dialects were of course not identical to the language spoken at that time (cf. Mattheier 2003: 214). Rather, they represent an approximation to the spoken language of that time.

with clear East Upper German features" (Mattheier 2003: 216–217). However, levelling processes were quite early interrupted by Luther's translation of the Bible and the Reformation in the 1520s and 1530s (cf. Mattheier 2003: 217). We therefore still consider it justified to assume that the East Upper German area and with it the written regional norm can be regarded as precursor of the Bavarian dialect, at least to a certain degree. Finally, we suggest that East Central German – with Leipzig, Jena, and Wittenberg as principal writing/printing centers (see Figure 1) – represents the precursor of written standard German. The writings of Martin Luther (1483–1546) and, more generally, writings in the context of Reformation, were based on the East Central German written norm and "acted as linguistic model of enormous prestige" (Mattheier 2003: 217). *Lutherdeutsch* 'Luther German' formed the basis of the generally acceptable German written standard norm; at the end of the 18^{th} century the East Central German norm was finally codified in the influential grammar of German by Gottsched (cf. Mattheier 2003: 217–218).[5]

2.4 Data

We have figured out three time periods (see Figure 2), which should allow us to track a possible (actuation of) change in language.

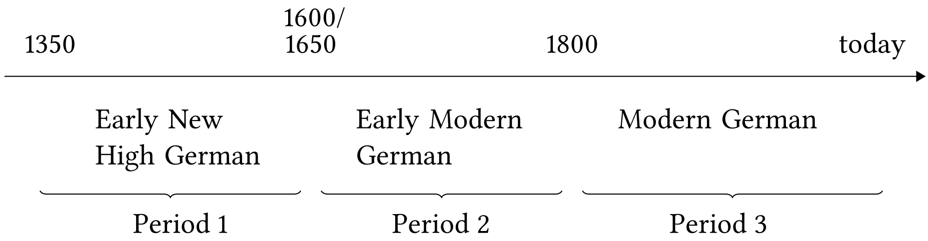

Figure 2: Periods of German

We will use both data from a corpus study (see Moser 2023, 2024) and from the literature: The data for our first time period (1350–1600) comes from the "Bonner Frühneuhochdeutschkorpus" (BFC), the data for the second period (1650–1800) from the "GerManC-Korpus" (GMC). Both corpora are described in detail in Moser (2023: 465–467, 2024). The total number of tokens for the first period is 48,000 (24 texts), for the second period a total of 120,000 (60 texts). These

[5]The standardization process in German was much more complicated as presented here; a good overview (in English) can be found in Mattheier (2003: in particular pp. 214–218).

differences are due to the different size of the two corpora (see for a detailed discussion, Moser 2023: 456–467, 2024). We have extrapolated the results to make them easier to compare: the number of texts was extrapolated to 48,000 words (or 24 texts) per dialect, see Table 1, and the number of relative markers was extrapolated accordingly (see the results in Section 4.1).

Table 1: Number of text extracts (original/extrapolated)

	West Upper G.	East Upper G.	East Central G.
Period 1	12	6	6
Period 1, extrapolated	24	24	24
Period 2	24	24	12
Period 2, extrapolated			24

The data for the third period (1800 onwards) is taken from the literature: The data on Alemannic (Swiss German and Alsatian Alemannic) comes from Dalcher (1963), Weise (1916) and the *Elsässisches Wörterbuch* (Alsactian Dictionary) (EWb 1899–1907),[6] that on Central Bavarian from Pittner (1996) and Bayer (1984), and that on written standard German from Eisenberg (2020).

3 The Noun Phrase Accessibility Hierarchy (AH)

The Noun Phrase Accessibility Hierarchy (AH) by Keenan & Comrie (1977) mirrors the accessibility of different syntactic positions to relativization, such that the leftmost position (subject) is easiest to relativize while the rightmost position is most difficult. When introducing the AH, we will focus on the following questions: What does the AH say, where might there be ambiguities in the formulation(s), and which (modified) version of the AH are we going to use?

3.1 The Hierarchy Constraints

The AH was first formulated in 1972 by Keenan & Comrie, and the published version from 1977 therefore represents a developed version of this earlier formulation. The authors attempt to determine the universal properties of relative

[6]In Swiss German and Alsatian Alemannic *wo* as relative marker is obligatory (cf. Bräuning 2020: 107).

clause (RC) formation;[7] their considerations are based on a large-scale investigation of data from 48 genetically unrelated languages spoken world-wide. The data was later published in Keenan & Comrie (1979) (although it was intended to be published as an appendix to the article from 1977). The universal principles (= hierarchy constraints) are represented in a hierarchy, which predicts that the further a position is located to the left, the easier it is to relativize (cf. Keenan & Comrie 1977: 88–89). The syntactic positions of the hierarchy are read from the right to the left, with ">" meaning "more accessible than"; e.g., the subject position is easier (or: more accessible) to relativize than the genitive position:

(4) Subject (SU) > direct object (DO) > indirect object (IO) > oblique (OBL) > genitive GEN) > object of comparison (OCOMP)
(Keenan & Comrie 1977: 66)

The Hierarchy Constraints (HC) are universal and define the following conditions (Keenan & Comrie 1977: 67):

1. A language must be able to relativize subjects.
2. Any RC-forming strategy must apply to a continuous segment of the AH.
3. Strategies that apply at one point of the AH may in principle cease to apply at any lower point.

As mentioned above, there are two different formulations of the AH, which differ as regards some important points: In their earliest formulation Keenan & Comrie propose that if a language can relativize on any low position of the AH it can relativize all the way up (cf. Keenan & Comrie 1977: 68). Because of certain counterexamples in some Malayo-Polynesian languages (Keenan & Comrie 1977: 68–69) this was then replaced with the continuous segment prediction (see HC 2). This prediction was defined on different "strategies" in the 1977 version, which no longer makes the "if low then high" prediction except for RC that include relativizations on subjects.

[7] Keenan & Comrie (1977) count as RCs both the "traditional" adnominal RC and participal constructions. In this article, we restrict ourselves to adnominal RCs. According to Fleischer (2004: 214) participal constructions do not exist (or very rarely) in German dialects.

3.2 Syntactic positions

Fleischer (2004: 215–216) discusses the syntactic positions in relation to German and its dialects: the SU position is encoded by the nominative, the DO position by the accusative, and the IO position by the dative. Fleischer (2004: 216) mentions that it is very difficult to find data for the IO position in corpora since this position has a very low frequency. This may explain the low number of evidence of relative markers in IO position in our data (see Section 4.1). Keenan & Comrie (1977: 72) also point out that the IO position is "the most subtle one on the AH" because in many languages it either assimilates IOs to DOs or to OBL cases.[8] OBL case only includes arguments of the main predicate, as the chest in *John put the money in the chest* (Keenan & Comrie 1977: 66). OBL case is usually introduced by a prepositional phrase (in German, cf. Fleischer 2004: 215, but also in English, cf. Herrmann 2001: 128–129, who therefore renames this position into "prepositional complement"). RCs having a more adverbial function like *Chicago* in *John lives in Chicago* with a locative reading ("where") or that day in *John left on that day* with a temporal reading ("when") are not seen as core members of OBL case (cf. Keenan & Comrie 1977: 66). However, they do not categorically exclude prepositional adverbials as oblique cases, too: In Keenan & Hawkins (1987: 63–65) this position is exemplified, among others, with optional prepositional complements functioning as adverbials. Both Herrmann (2001) and Fleischer (2004) only consider prepositional complements for the OBL position, following thus Keenan & Comrie's narrow definition from 1977. In our study, too we will include OBL as understood by Herrmann (2001) and Fleischer (2004). But in addition to that, we will add another category, which includes RC as adverbial complements or adjuncts (= obligatory and optional adverbials) with a locative reading (LOC) (cf. in a similar vein, Lehmann 1984: 668, who distinguishes between local, temporal, and other complements as well as adjuncts). Note that the AH and with it the continuous segment prediction only holds for SU > DO > IO > OBL as suggested (and exemplified with cross-linguistic data) in Keenan & Comrie (1977, 1979). Finally, we follow Fleischer (2004) and leave out both GEN and OCOMP position: The GEN case "is virtually non-existent in almost all German dialects" (Fleischer 2004: 216) and OCOMP does exist neither in Standard German (cf. Keenan & Comrie 1977: 74) nor in its dialects. Our slightly modified version of Keenan & Comrie's AH 1977 now looks as follows:

(5) SU > DO > IO > OBL | LOC

[8] In Low German dialects, for example, accusative and dative merged in a great number of instances (cf. Fleischer 2004: 215).

3.3 RC-forming strategies

Strategies were not defined ad-hoc for each language, but they should be distinguished on universal grounds. Keenan & Comrie (1977: 64) observe that within a language there might be more than one strategy, and that "[d]ifferent strategies differ with regard to which NP positions they can relativize". The authors define "strategy" in terms of a combination of the ordering of head noun plus RC (e.g., postnominal relatives vs. pronominal ones) and of so-called "case-coding" (e.g., with resumptive pronouns in the relative clause vs. gaps) (cf. Keenan & Comrie 1977: 63–66). Technically, Keenan & Comries' sense of strategy does not define RCs with *wo* or with other particles in our German data as distinct strategies. When they illustrate the continuous segment prediction with their cross-linguistic data, however, their term "strategy" is defined quite differently: In Swedish, for example, two postnominal strategies co-exist, namely one involving pronominalization and the other one involving relative particles (Keenan & Comrie 1977: 79). Note that a resumptive pronoun or a preposition in the restricting clause, in addition to a relative marker, counts as [+ case RC strategy]: clauses such as *the chest in which* $_{[OBL]}$ *John put the money* are therefore considered [+ case RC strategy] (cf. Keenan & Comrie 1977: 66). Another, similar proposal for the classification of RC-forming strategies comes from Maxwell (1979): He suggests introducing a relative pronoun strategy (REL-S) and an anaphoric pronoun strategy (PRO-S) (among others) (cf. Maxwell 1979: 358). To put it very simply, in a relative pronoun strategy, relative pronouns encode case and introduce a RC (cf. Maxwell 1979: 356–357). An anaphoric pronoun strategy, on the other hand, does not introduce the subordinate clause (cf. Maxwell 1979: 358): "[T]his strategy might therefore also be called retention strategy; the pronoun encoding case is often referred to as the resumptive pronoun" (Fleischer 2004: 228). In the following, we will use the term "strategy" as Keenan & Comrie use it when illustrating the continuous segment prediction with their data (Keenan & Comrie 1977: 76–79, Keenan & Comrie 1977). Fleischer (2004) (for German dialects) and Romaine (1984) (for Germanic varieties) adopt this interpretation of the term "strategy" as well. We will not use the term "strategy" in the sense of "one form is one strategy" as this certainly was not the intention of Keenan/Comrie.[9] Strategies were not defined ad-hoc for each language, but they should be distinguished on universal grounds. Keenan & Comrie (1977: 64) observe that within a language there might be more than one strategy, and that "[d]ifferent strategies differ with regard to which NP positions they can relativize". The authors define "strategy" in

[9] This interpretation is used in Herrmann (2001: 127–147) where relativization with *what* and *as* are described as different RC-forming strategies in English dialects.

terms of a combination of the ordering of head noun plus RC (e.g., postnominal relatives vs. pronominal ones) and of so-called "case-coding" (e.g., with resumptive pronouns in the relative clause vs. gaps) (cf. Keenan & Comrie 1977: 63–66). Technically, Keenan & Comries' sense of strategy does not define RCs with *wo* or with other particles in our German data as distinct strategies. When they illustrate the continuous segment prediction with their cross-linguistic data, however, their term "strategy" is defined quite differently: In Swedish, for example, two postnominal strategies co-exist, namely one involving pronominalization and the other one involving relative particles (Keenan & Comrie 1977: 79). Note that a resumptive pronoun or a preposition in the restricting clause, in addition to a relative marker, counts as [+ case RC strategy]: clauses such as *the chest* <u>*in which*</u> $_{[OBL]}$ *John put the money* are therefore considered [+ case RC strategy] (cf. Keenan & Comrie 1977: 66). Another, similar proposal for the classification of RC-forming strategies comes from Maxwell (1979): He suggests introducing a relative pronoun strategy (REL-S) and an anaphoric pronoun strategy (PRO-S) (among others) (cf. Maxwell 1979: 358). To put it very simply, in a relative pronoun strategy, relative pronouns encode case and introduce a RC (cf. Maxwell 1979: 356–357). An anaphoric pronoun strategy, on the other hand, does not introduce the subordinate clause (cf. Maxwell 1979: 358): "[T]his strategy might therefore also be called retention strategy; the pronoun encoding case is often referred to as the resumptive pronoun" (Fleischer 2004: 228). In the following, we will use the term "strategy" as Keenan & Comrie use it when illustrating the continuous segment prediction with their data (Keenan & Comrie 1977: 76–79, Keenan & Comrie 1977). Fleischer (2004) (for German dialects) and Romaine (1984) (for Germanic varieties) adopt this interpretation of the term "strategy" as well. We will not use the term "strategy" in the sense of "one form is one strategy" as this certainly was not the intention of Keenan/Comrie.[10]

3.4 Short summary

Our basic distinction is the [+/– case RC strategy], which always appears postnominal (in German). The [+ case RC strategy] includes pronominalization (*d-* and *wh*-pronouns for German), while the [– case RC strategy] consists of an invariable particle (*da*, *so*, *wo* for German). Relative clauses including a resumptive pronoun (in addition to a relative particle) or a preposition (such as in OBL position) are considered as [+ case strategy] as well.

[10]This interpretation is used in Herrmann (2001: 127–147) where relativization with *what* and *as* are described as different RC-forming strategies in English dialects.

4 RCs in the (historical) dialects

4.1 RC-forming strategies

Tables 1–3 were created using Keenan & Comrie (1977: 76–79) as a model. The numbers in the cells represent the number of relative markers for the respective position and strategy. In Early New High German (1350–1600), two different strategies in SU and DO position are attested: [+ case RC strategy] and [– case RC strategy] (see Table 2).

Table 2: Distribution of RC-forming strategy, period 1

Strategy	Relativizable positions			
	SU	DO	IO	OBL
West Upper German				
– case	46	18		
+ case	358	80	12	46
East Upper German				
– case	28	12		
+ case	448	156		32
East Central German				
– case	32			
+ case	468	88		20

The [– case RC strategy] is realized with so in SU and DO position; in the following example DO position is relativized:

(6) East Upper German (1557)
aber vor etlich jarn/ hat der Tůrckh Weissenburg/ **so** man auch
but before several years has the Turk Weißenburg which one also
Moncastro nent/[…] ein genumen
Moncastro calls taken over
'but several years ago the Turk took over Weißenburg which we also call Moncastro'
Sigmund Freiherr von Herberstein "Moskau" (F038–003,31)

The [+ case RC strategy] is realized in the form of an inflected *d*-pronoun, from SU position down to OBL case. There is a decrease in frequency from the higher to the lower positions, just as the AH would predict. The IO position is so far only attested in West Upper German. The pronominalization strategy is somewhat less frequent in West Upper German than in the other dialects, but here we find the highest number of relative markers using a particle strategy.

In early Modern (New High) German dialects (1650–1800), the same two RC-strategies in the same positions are attested (see Table 3). Pronominalization strategies are now possible in IO position in all three dialects as well while the particle strategy is still only used in the upper two positions of the AH.

Table 3: Distribution of RC-forming strategy, period 2

Strategy	Relativizable positions			
	SU	DO	IO	OBL
West Upper German				
− case	16	6		
+ case	361	95	13	78
East Upper German				
− case	22	10		
+ case	326	71	6	44
East Central German				
− case	32	8		
+ case	356	100	12	64

As for pronominalization, a new marker has appeared: *wh*-pronoun is, just like *d*-pronoun, possible in all positions from SU to OBL.[11] The following example shows OBL case with a *wh*-pronoun:

(7) West Upper German (1798)

Briefe aus Bordeaux sind des Růhmens voll ůber den Eifer,
letters from Bordeaux are of-the praise full about the eagerness
mit welchem die dortige Bůrgerschaft es der Pariser in ihrem
with which the there citizens it the Parisians in there

[11]There are basically three different scenarios for the origin of the *wh*-pronoun in German (cf. Moser 2024): language contact to Middle Low German, the reanalysis of ambiguous interrogative pronouns, or a calque modeled on Latin and/or French.

Eifer gleichzuthun sucht
eagerness imitate search
'letters from Bordeaux are full of praise about the eagerness with which the citizens there are trying to imitate the Parisians in their eagerness'
Neueste WeltKunde (Tübingen 1798: vol. 1/14)

And in this example, the IO position is realized with a *d*-pronoun:

(8) West Upper German (1556)
Der gůt herr **dem** diss gar ein rauhe sach war/
the good lord whom this quite a rough affair was
'the good gentleman, for whom this was quite a rough affair'
Hans Georg Wickram "Von Guten und Bosen Nachbaur" (F082- 902r,13)

As already in period 1, the [– case RC strategy] covers SU and DO position and is realized with so. This strategy is now most frequently used in East Upper and East Central German. The following example shows so, relativizing the subject position of the relative clause:

(9) East Central German (1722)
Aus Aachen wird unterm 10 dieses gemeldet, daß den 6 Mr. de St.
from Aachen is under 10 this reported that the 6 mister de saint
Distant,[…], mit einem magnifiquen Geschenck vor dasige Stiffts-
distanz with a beautiful gift for this collegiate
Kirche, **so** in dem kostbaren Leichen-Tuch des verstorbenen Kǒnig
church which in the precious shroud of-the deceased king
Ludwigs des XIV. bestanden,[…] angelanget sey
Ludwig the XIV. consisted reached was
'It is reported from Aachen on the 10th that a beautiful gift for this collegiate church, consisting of a precious shroud of the deceased King Ludwig, reached the 6th Mr. de Distant'
Postzeitungen (Leipzig 1722)

In modern dialects and written standard German, the two parallel strategies in SU and DO position have been replaced by one single or two strategies. In the case of two strategies, each strategy covers a continuous segment of the AH, without any overlap with the other strategy (see Table 4).

Table 4: Distribution of RC-forming strategy, period 3

Strategy	Relativizable positions			
	SU	DO	IO	OBL
Alemannic				
- case	+	+	−	−
+ case	−	−	+	+
Bavarian				
+ case	+	+	+	+
Written st. German				
+ case	+	+	+	+

In written standard German, only [+ case RC strategy] is possible, either with a *d-* or *wh*-pronoun (cf. Eisenberg 2020: 296). In Alemannic, two strategies are attested: SU and DO position are relativized with *wo* [− case RC strategy]. The following example shows *wo*, relativizing DO position (for an example of *wo* in SU position, see Section 2.1):

(10) Alemannic
dä Wäg, **wo** du im Sinn hesch
the path where you in mind have
'the path which you have in mind'
Weise (1916: 66)

IO position and OBL case are relativized with *wo* and resumptive pronoun or preposition in the RC (cf. Dalcher 1963: 127–129), thus a [+ case RC strategy]. The following example shows *wo* plus resumptive pronoun, used in IO position:

(11) Alemannic
Die selben Hüener **wo-n-enen** d' Büben Fäderen usezerrt hän
the same chickens where-them the boys feathers plucked have
'The same chickens that the boys have plucked feathers from'
Dalcher (1963: 127; simplified characters)

And in this example, we find a [+ case RC strategy] in OBL case, with *wo* and preposition in the RC:

(12) Alemannic
Die Büecher, **wo** de mir s Gëld **derzue** gegëben hest co.
the books where you me the money to-it given have could
'the books you could have given me the money for'
EWb (1899–1907: II, 778)

In Bavarian, OBL case is realized in the form of *wo* plus a preposition inside the RC (the same pattern as in Alemannic, see example 12). The combination *wo* + *d*-pronoun is attested for SU, DO, and IO position. Note that the relative particle *wo* is not obligatory in SU, DO and IO position (cf. Pittner 1996: 135). At the same time, under certain conditions, the relative pronoun can be left out, such as when the relative pronoun has the same case as the head noun (cf. Pittner 1996: 135–136, Bayer 1984: 216): The following example shows *d*-pronoun + *wo*, relativizing the DO position of the RC:

(13) Central Bavarian
Der Mantl **den** **wo** i kaffd hob wor z'rissn
the coat which where I bought have was torn
'The coat which I bought was torn'
Pittner (1996: 135, after Bayer 1984: 216)

4.2 Discussion of the results

4.2.1 In the light of Fleischer (2004) and Romaine (1984)

In our historical data [– case RC strategy] is only possible in SU and DO position while [+ case RC strategy] is attested from SU position down to OBL case. According to Fleischer (2004)'s study on German modern dialects, there is a separation of strategies between SU and DO position and OBL case (with IO position with the status of a changeable intermediate position). With regard to our historical data, this is not the case, or only to a very limited extent: Although the particle strategy is only found in SU and DO position, the pronominalization strategy is used in SU and DO position as well. OBL case, on the other hand, is relativized by one strategy. With respect to this position, one could therefore speak of a separation of strategies.

Based on a (diachronic) study on Germanic languages and English-based pidgins/creoles, Romaine (1984) suggests that the AH represents a continuum with (at least) three well-defined points (1984: 445). Note that each point corresponds to a RC-forming strategy. RCs with invariant relative particles and relativization strategies involving pronominalization mark the two end points. The third point and type is described as "movement along the continuum […] in which two strategies co-exist" (Romaine 1984: 445). According to her approach, this would mean that – from a synchronic perspective – Alemannic is in the middle of the continuum (two strategies), while written German and Bavarian each mark an end point, namely the pronominalization one (= [+ case RC strategy]). In diachronic terms, two of the three historical dialects would have moved to an end point on the continuum while the third dialect would still be between the two end points. In the following, we would like to propose a novel interpretation of the AH, which in our opinion allows us to explain the shift or change in strategy in our data.

4.2.2 A novel interpretation of the AH

Based on our data, the following diachronic changes can be made with regard to RC-forming strategies: In periods 1 and 2, there is an overlap in strategy since the two highest positions (SU and DO) are occupied twice, by two strategies for the same position. In period 3, this "double strategy" is replaced by either one strategy (Bavarian and written standard German) or by two strategies (Alemannic). In the case of two strategies, however, these are distributed over two different but continuous segments, without any overlap.

We now propose that the AH can serve as an explanation for this change in the RC-forming strategy. We argue that the distribution of strategies, as found in period 3, is typologically expected while the distribution of strategies, as found in periods 1 and 2, is typologically unusual. If we look at the AH as presented in the table with cross-linguistic data by Keenan & Comrie (1977: 76–79), we see that there are 28 languages with two postnominal strategies, namely a [+ case RC strategy] and a [– case RC strategy]. 24 of them have no overlaps in strategy, i.e., no two strategies for the same position. There is only evidence for overlaps in strategy in 4 languages (Genoese, Hebrew, Persian, Tongan). Otherwise, it is always the case that one strategy ends (e.g. in DO position) and the other strategy starts at a different position (e.g. in IO position). This means that in most languages the continuous segment prediction is spread over two different segments.[12] The historical dialects, however, do not behave like most languages with two postnominal strategies, and that is why we call the distribution of strategies, as found in periods 1 and 2, typologically unusual. So, one could argue that language change, as it has occurred in German in terms of its RC-forming strategies

[12]This is true even if all languages, not only those with two postnominal strategies, are included: In this case, 27 languages have 2 strategies (but are spread over different segments), and 8 languages have two strategies with overlap.

([+/− case RC strategy]), is to some extent predicted by the AH or continuous segment prediction.

5 Diachronic explanations

The continuous segment prediction, as interpreted in Section 4.2, says that more than one strategy for the same position(s) of the AH is typologically unusual (but not impossible). Changes in relation to the form of the relative marker, on the other hand, cannot be predicted by the AH. Instead, language-internal mechanisms and language-external factors must be considered.

5.1 Locative (and temporal) *wo*

In the literature *wo* in a locative function (LOC) is first – and only sporadically – attested at the end of the 15th or beginning of the 16th century (cf. Moser 2023: 473, Behaghel 1928: 732–733).[13] Until the 17th century, the function LOC is mainly relativized with the (polyfunctional) particle *da* (cf. Moser 2023: 473). The following example shows the invariable particle *da* relativizing the head noun *vff den schouwplatz*:

(14) West Upper German (1578)

der	sye	vil	tag	vff	den	schouwplatz	gangen/	**da**	mā
this-one	such	a-lot	day	on	the	stage	went	where	one

sunst	pflågt	Comedien	zů	spil̄ē/
otherwise	used	comedies	to	play

‘this one spent so many days on the stage where comedies are usually performed’
Ludwig Lavater “Von Gespansten vnghüren” (F106–012r,21)

From the 17th century onwards, locative *da* is very probably getting replaced by *wo* since it appears more frequently at exactly the moment (second half of 17th century) when *da* disappears (cf. Moser 2023: 472–476). The following example shows *wo* with a locative reading:

[13] Before that, it is used as interrogative pronoun (cf., e.g. Paul 2007: 223, §223).

(15) East Upper German (1798)
Wie in England, wo man, ohngeachtet der strengen
as in England where one despite the strict
ZollAnstalten, ganze Låden voll verbotener Waaren in den Straßen
customs-regulations entire shops full prohibited goods in the streets
sieht
see
'As in England, where, despite the strict customs regulations, you can see whole shops full of forbidden goods in the streets'
Neueste Weltkunde Tübingen (Zeitung)

Wo can be used with a temporal reading as well (this was also possible for *da*, cf. Moser 2023: 474–475 for further details), but this function is less frequent than *wo* with a locative function (only 9 pieces of evidence for temporal function vs. 67 pieces of evidence for LOC function in our data). The following example shows *wo* with a temporal function in the RC, relativizing the PP *to his empty future*:

(16) East Upper German (1796)
[…] håtte nicht Thienette […] die Sieche auf den armen Konrektor
had not Thienette the sieche to the poor dean
gebracht […] und auf seine leere Zukunft, **wo** er als gelbes mattes
brought and to his empty future where he as yellow dull
Gewåchs in den trockenen Dielen-Fugen der Schulstube
plant in the dry joints-of-the-floorboards the school
zwischen Schůlern und Glåubigern welken werde.
between pupils and creditors wither would
'(if) Thienette hadn't brought this illness to the poor dean and to his empty future, where he would wither as a dull yellow plant in the dry floorboards of the school between pupils and creditors'
Jean Paul "Leben des Quintus Fixlein" (1796: ch. 5, pp. 102–108)

In written standard German and the modern dialects, *wo* can be used to relativize a NP with locative and temporal semantics (cf. Eisenberg 2020: 300). Although the temporal use of *wo* is seen as an innovation in German, its use is as grammatical as the one of local *wo* and is accepted by grammarians since the 19th century (cf. Davies & Langer 2006: 129). However, locative *wo* is actually used much more frequently than temporal *wo* (cf. Davies & Langer 2006: 124–139).

5.2 Relative *wo* (SU, DO position)

The earliest evidence of *wo* in SU position can be found in the 16th century in a theological sermon-like work by Thomas Murner, printed in Strasbourg (cf. Dalcher 1963: 121):

(17) West Upper German (1514)
wir kumment widerum zü got: ja **wo** uns unser sünden lot
we come again to god PART where us our sins forgives
'We come back again to God, who forgives us our sins.'
Murner (1514, cited after: DWb 30: c. 916)

One century later, we have evidence of *wo* in DO position:

(18) West Upper German (1641)
han den grosen Oxen auch gschauwet, **wo** s gmetzget hand
have the big ox also seen which they slaughtered have
'I also saw the big ox they slaughtered'
Dalcher (1963: 121)

Further evidence, this time for the SU position again, can be found almost a century later:

(19) West Upper German (1703)
der **wo** anderswo Burger ist worden, verliert sein Burgerrecht
anyone who elsewhere citizen is become loses his citizenship
'Anyone who has become a citizen elsewhere loses his citizenship'
Council minutes of the village Mühlhausen, from December 1703, cited after: EWb: 779 2)

Note that this evidence (examples 17 to 19) is only attested for West Upper German. In 1760 a scholar from Basel mentions for Alemannic that *wo* is often misused for *d*-pronoun (cf. Dalcher 1963: 116). In the 19th century, *wo* in SU and DO position is found with authors of Alemannic-speaking origin (cf. DWb 30: c. 916). And in the 20th century, Fischer (1960: 429) observes that the relative pronoun has been lost in Alemannic and that the invariable particle *wo* is used instead (in a similar vein, cf. Weise 1916: 67). There are various considerations as to why we find *wo* in SU and DO position in Alemannic today. It may have something to do with the fact that Alemannic was hardly exposed to any standardization processes and was therefore able to develop "in a natural way" (see

Section 2.3, Moser 2023: 479–482). *Wo* could then have become grammaticalized, from a marker with locative/temporal function into a general relative clause marker (see Moser 2023: 476–479).[14] It is equally possible that the spread of *wo* in SU and DO position has to do with *so*: In our data, evidence for *so* in SU and DO position is decreasing since the 17th century, and *so* is rarely attested after 1800 in Alemannic (cf. Dalcher 1963: 117, for Swiss German). It could therefore have been possible that *wo* replaced *so* as relative marker: in contrast to *wo*, *so* was incompatible with prepositions (thus not possible in OBL case) (cf. Reichmann & Wegera 1993: 447); what is more, *so* was limited to one specific register (*Kanzleisprache*), and this register became unpopular in the 18th century (cf. Pickl 2020: 251–253). It could therefore be argued that *wo* was more suitable both in structural terms (more functions possible) and in stylistic terms, at least as a relative marker in a *spoken* language variety.

5.3 *D*-pronouns in written standard German

In written standard languages such as written standard German, "universal factors of grammatical explicitness and unambiguity" (Pickl 2020: 254) speak in favor of relative pronouns and against relative particles. Interestingly, and from a (European) typological perspective unexpected, *d*-pronouns – and not *wh*-pronouns – are the preferred relative marker in German.[15] In functional-structural terms, *wh*-pronouns are on a par with *d*-pronouns in German as they are possible in the same functions (SU, DO, IO, OBL), and can be used in locative function as well:

(20) written standard German

a. Das ist die Stadt, **wo** ich wohne
   this is the town where I live

b. Dsa ist die Stadt, **in der/welcher** ich wohne
   this is the town in which I live

"The town where I live"
After Eisenberg (2020: 300)

[14] Brandner & Bräuning (2013), on the other hand, suggest that relative *wo* in Alemannic evolved out of equative *so*. Their analysis is underlined with evidence from Scandinavian varieties, in which the relative particle *som* goes back to an equative particle, which is historically directly related to *so*.

[15] In most European languages, "the relative pronoun is based on an interrogative pronoun" (Haspelmath 2001: 1419). In German, however, the relative pronoun is based on a demonstrative.

Wh-pronouns are attested from the 16th century onwards, in all writing landscapes and with increasing frequency (cf. Moser 2024, in a similar vein: Pickl 2020). East Central German – as the forerunner of the written standard (cf. Mattheier 2003) – did not have a special role to play here, as all writing/printing landscapes behaved in a similar way (cf. Moser 2024). After a high flight in the 18th century, where *wh*-pronouns were even used more frequently than *d*-pronouns in some text genres (cf. Pickl 2020), its use declined sharply. According to Pickl (2020: 251–252), this was due to both normative and stylistic reasons: While grammarians preferred the use of *wh*-pronouns in the 18th century, *d*-pronouns were preferred in the 19th century (and *wh*-pronouns were stigmatized). What is more, *d*-pronouns were better anchored in spoken (dialectal) language while *wh*-pronouns were limited to written language (cf. Fleischer 2005: 185, Brooks 2006: 135). Even today, *wh*-pronouns are considered stylistically cumbersome and unattractive and are only rarely used (cf. Eisenberg 2020: 296).

5.4 *D*-pronoun + *wo* in Bavarian

There is no evidence of a combination of *wo* with relative pronoun until recently (e.g., neither mentioned by *Deutsches Wörterbuch von Jacob Grimm und Wilhelm Grimm* (DWb), by Weise (1916) nor in my data), which suggests that this construction is of more recent date and must have evolved at some point in the 20th century. From a synchronic perspective, the combination *d*-pronoun + *wo* represents an innovation. From a diachronic perspective, however, this pattern appears to be more the result of a substitution process (cf. Fleischer 2004: 232): A very similar combination (*d*-pronoun + *da*) is already attested in earlier stages of German (cf. Moser 2024) and is still found in the 18th century in High German dialects (cf. Brooks 2006: 124–125). Today, this pattern is still attested in Low German (cf. Fleischer 2004). Furthermore, the construction "*d*-pronoun + x" (with x representing an invariable particle) is already found in Old High German and in Middle High and Middle Low German (cf. Lehmann 1984: 379, Fleischer 2004: 235, 239). We therefore propose that *wo* replaced *da* in its function as locative relative in the historical High German dialects (see. Section 5.1), and then, in the same way, replaced the particle *da* with *wo*, but only in Bavarian. The combination in Bavarian is therefore both an innovation and the result of a substitution process.

6 Summary

We compared relative clause (RC)-forming strategies, as understood by Keenan & Comrie (1977), in Upper German dialects (Bavarian and Alemannic) and written standard German with those in the historical dialects of earlier time stages (1350–1800). In the historical dialects (1350–1800, periods 1 and 2) relativization is possible in subject (SU) and direct object (DO) position with two different RC-forming strategies at the same time ([+ case and − case RC strategy]). In the modern dialects (including the written standard language) (period 3) SU and DO position can only be relativized with either a [+ case RC strategy] or a [− case RC strategy]. We proposed that the distribution of strategies, as found in period 3, is typologically expected while the distribution of strategies, as found in periods 1 and 2, is typologically unusual. Language change, as it has occurred in German in terms of its RC-forming strategies ([+/− case RC strategy]), is therefore to some extent predicted by the AH or continuous segment prediction: The continuous segment prediction, as exemplified in Keenan & Comrie's cross-linguistic data, says that more than one strategy for the same position(s) of the AH is typologically unusual (but not impossible). However, the AH cannot make any statements about very specific changes, e.g. on the form side. In this case, language-internal mechanisms and language-external processes such as grammaticalization, standardization, innovation and substitution come into play.

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Acting on actuation

This volume presents a timely discussion on one of the most fundamental and yet elusive questions in historical linguistics: why do certain linguistic changes take place in some languages at specific times, but not in others, even under similar conditions? The so-called actuation problem, first articulated by Weinreich, Labov, and Herzog (1968), remains a central puzzle in the study of language change, at the crossroads between language structure, cognitive processes, and social dynamics. While significant progress has been made in identifying pathways and constraints on change and in understanding the social embedding of linguistic variation, the ultimate challenge of predicting language change remains unresolved, raising the question of whether historical linguistics can ever be a predictive science. The main reason for skepticism is that the inherent complexity of language structure and use makes it extremely challenging to predict when and how a given change may occur. Even so, a reassessment of where the discipline stands with respect to its most central research question is in order.

Building on recent advances in variationist sociolinguistics, grammaticalization theory, and probabilistic modeling of language, the contributions in this volume offer fresh theoretical and methodological perspectives on the actuation problem, discussing the interplay between principles of language change, the role of bilingualism and language contact more generally, the distinction between innovation and propagation, and the role of sociocultural change. Research presented in this volume shows that there is indeed cause for hope, bringing at least a probabilistic answer to the actuation problem within closer reach.